



ICT378 – Cyber Forensics & Incident Response Narcos 2019 Investigation Report

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TABLE OF CONTENTS

1. Executive Summary	4
1.1 Brief Description of the Event	4
1.2 Brief Methodology of the Investigation	4
1.3 Evidence Collection and Preservation Methods	5
1.4 Conclusion	5
2. Methodology Details	6
2.1 Investigation	6
2.1.1 Autopsy	6
2.1.2 Magnet AXIOM	7
2.1.3 ChromeCacheView	9
2.2 Evidence Collection and Preservation	9
3. Findings	10
3.1 Finding 1 – User Accounts & Computer Names	10
3.1.1 Discussion – Steve Kowhai	10
3.1.2 Supporting Evidence – Steve	10
3.1.3 Discussion – John Fredrickson	11
3.1.4 Supporting Evidence – John	11
3.1.5 Discussion – Jane Esteban	12
3.1.6 Supporting Evidence – Jane	12
3.2 Finding 2 – Roles of Each Suspect	13
3.2.1 Discussion – Steve Kowhai	13
3.2.2 Supporting Evidence – Steve	13
3.2.3 Discussion – John Fredrickson	15
3.2.4 Supporting Evidence – John	15
3.2.5 Discussion – Jane Esteban	16
3.2.6 Supporting Evidence – Jane	16
3.3 Finding 3 – Web Activity	17
3.3.1 Discussion – Steve Kowhai	17
3.3.2 Supporting Evidence – Steve	17
3.3.3 Discussion – John Fredrickson	17
3.3.4 Supporting Evidence – John	18
3.3.5 Discussion – Jane Esteban	19
3.3.6 Supporting Evidence – Jane	19
3.4 Finding 4 – Images Related to Behavior	20
3.4.1 Discussion – Steve Kowhai	20

3.4.2 Supporting Evidence – Steve	20
3.4.3 Discussion – John Fredrickson	22
3.4.4 Supporting Evidence – John	22
3.4.5 Discussion – Jane Esteban	23
3.4.6 Supporting Evidence – Jane	23
3.5 Finding 5 – Binary Files	24
3.5.1 Discussion – Steve Kowhai	24
3.5.2 Supporting Evidence – Steve	24
3.5.3 Discussion – John Fredrickson	24
3.5.4 Supporting Evidence – John	25
3.5.5 Discussion – Jane Esteban	25
3.5.6 Supporting Evidence – Jane	25
3.6 Finding 6 – Communications	25
3.6.1 Discussion – Steve & John	25
3.6.2 Supporting Evidence – Steve & John	26
3.6.3 Discussion – Jane Esteban	26
3.6.4 Supporting Evidence – Jane	26
3.7 Finding 7 – Documents	27
3.8 Finding 8 – Obfuscation Methods	29
3.9 Finding 9 – Encryption Methods	30
3.10 Finding 10 – Malware	33
3.11 Finding 11 – Vulnerability	34
3.12 Finding 12 – Windows Artefacts (Timeline)	35
3.13 Finding 13 – OS Build Differences	37
4. Conclusion	39
4.1 Summary of Findings	39
4.2 Evidence of Illegal Drug Activity	39
4.3 Reliability of Evidence Collection	39
4.4 Innocent / Guilty Decision	39
5. Appendix	40
5.1 Persons of Interest	40
5.2 Association Diagram	40
5.3 Table of Findings	41
5.4 Evidence Listing	41
5.5 Evidence Timeline	41
5.6 Software & Tools Used	42
5.7 References	42

1. EXECUTIVE SUMMARY

1.1. BRIEF DESCRIPTION OF THE EVENT

In 2019, New Zealand Customs caught two passengers, **John Fredricksen** and **Jane Esteban**, who were suspected of being involved in drug-related activities. Both individuals arrived from Brisbane. Officers located Windows laptop inside their suitcases. After further investigation of Jane's suitcase, they found 1kg of methamphetamine hidden inside the suitcase, which raised suspicion of organised smuggling.

Both were detained for questioning. John refused to cooperate, while Jane stated that she had been told to deliver the suitcase to Eastbourne Library, with a backup location at 666 Rewera Avenue, Petone, in case the meeting failed. Following this information, officers raided the address and found drugs, firearms, and a desktop computer, although no one was present at the time.

This report presents the findings from **three devices**:

- **Narcos-1:** Steve's desktop
- **Narcos-2:** John's laptop
- **Narcos-3:** Jane's laptop

1.2. BRIEF METHODOLOGY OF THE INVESTIGATION

The devices and memory dumps were handed over in a controlled manner. Because each device ran a different Windows 10 build, artefacts had to be examined carefully and individually.

All evidence discovered in this process is documented in the findings section of this report, supported by screenshots, file paths, and hash values.

1.3. BRIEF EVIDENCE COLLECTION AND PRESERVATION METHODS

Evidence collection followed standard forensic procedures to ensure that no data was altered or contaminated. The process included:

- **Securing the physical devices**

All three devices were stored in a controlled environment to prevent tampering, damage, or data modification.

- **Hash Verification**

MD5 hashing was applied to every drive image and memory dump to verify integrity. This ensured any extracted data could be traced back to an unchanged original.

- **Secure Evidence Storage**

Extracted artefacts, reports, and screenshots were stored in an access-controlled location.

This process ensured that the investigation remained reliable, repeatable, and compliant with recognized forensic standards.

1.4. CONCLUSION

The forensic investigation confirmed that all three devices contained digital evidence connected to the smuggling operation. By following proper forensic procedures and using trusted tools, we can confirm that the findings remain admissible and credible. Key requirements that were prioritised included:

- **Maintaining controlled storage conditions** to prevent damage or data loss.
- **Ensuring originality** by examining only forensic copies while preserving the original drives.
- **Maintaining full access logs and chain-of-custody records** to track every interaction with the evidence.

The findings show strong evidence against **John** and **Steve**, while **Jane** appears to have been acting under an undercover role and under direct threats.

2. METHODOLOGY DETAILS

2.1. INVESTIGATION

The investigation involved analysing three separate folders: **Narcos1** (Steve Kowhai), **Narcos2** (John Fredricksen), and **Narcos3** (Jane Esteban). Each folder contained a full disk image and a memory dump from one device (one desktop and two Windows laptops). These datasets were examined using a combination of strong forensic tools to extract artefacts, recover deleted items, identify communications, etc.

All tools used in this investigation were selected based on their reliability, efficiency, and suitability for the types of artefacts present in this case.

2.1.1. AUTOPSY

Autopsy was the main tool used during the investigation due to its flexibility, transparency, and strong performance in file system analysis (Morris & Choo, 2021). The reasons for using Autopsy for this forensic investigation include:

- **Cost-Effectiveness:**
Autopsy is open-source, making it suitable for detailed investigations without licensing restrictions.
- **Extensibility:**
The tool supports a wide range of community-driven modules, allowing investigators to customise the analysis based on the artefacts present.
- **Core Forensic Capabilities:**
Autopsy provided file system navigation, keyword searching, deleted file recovery, metadata inspection, and timeline reconstruction, all essential for understanding user behaviour across the three devices.
- **Detailed Manual Control:**
Autopsy allows investigators to manually check structures such as AppData folders, downloads directories, program data, and registry hives, providing precise insight into how each suspect used their device.

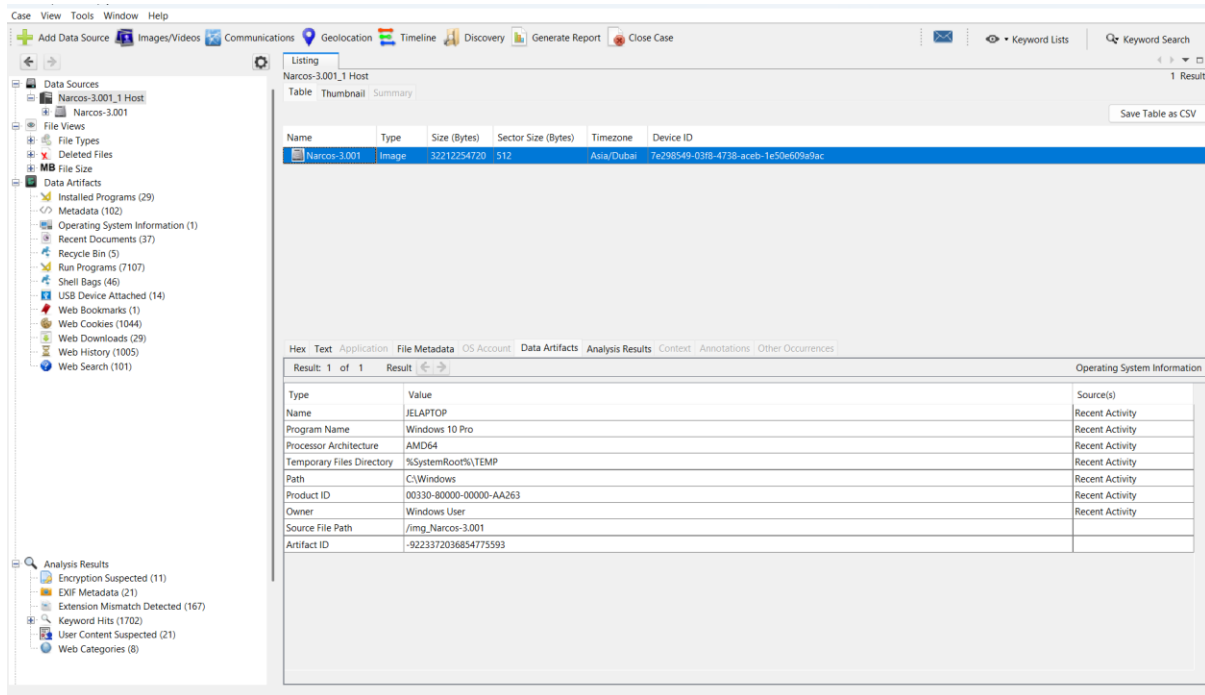


Figure 1: Autopsy

2.1.2. MAGNET AXIOM

Magnet AXIOM was also used to process and analyse the Narcos images. AXIOM was used for this investigation due to its strong artefact recovery capabilities, intuitive interface, and all-in-one design (Khan & Stanković, 2022). Its main advantages in this case include:

- **Advanced Artefact Recovery:**

AXIOM automatically extracted browser history, search terms, downloaded files, deleted items, images, communication remnants, and traces of malware activity.

- **Automation:**

AXIOM's carving and filtering capabilities helped locate important files quickly, such as the Quasar RAT installer, references to encrypted files, and certain browser queries related to undercover behaviour.

- **Visualisation Tools:**

The timeline and link analysis views helped connect activity across the devices, making it easier to understand relationships and communication patterns between John, Jane, and Steve.

- **Ease of Use:**

AXIOM's interface made it easier to combine artefacts and view them in context, especially when reviewing web activity and user actions.

Although Autopsy served as the main investigation tool, AXIOM helped verify findings, strengthen observations, and speed up the artefact extraction process.

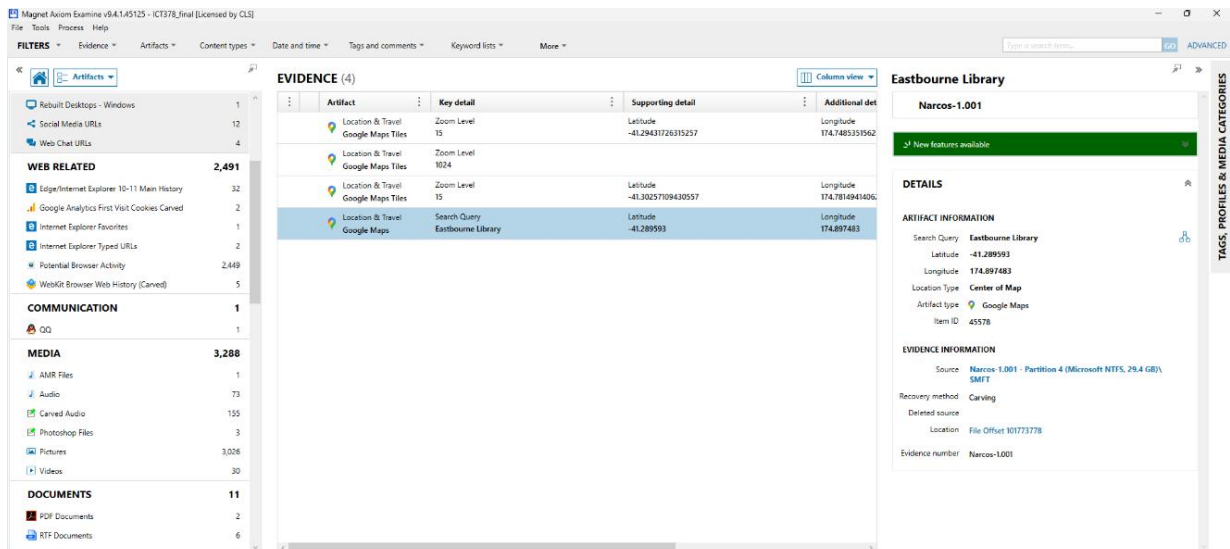


Figure 2: Magnet Axiom

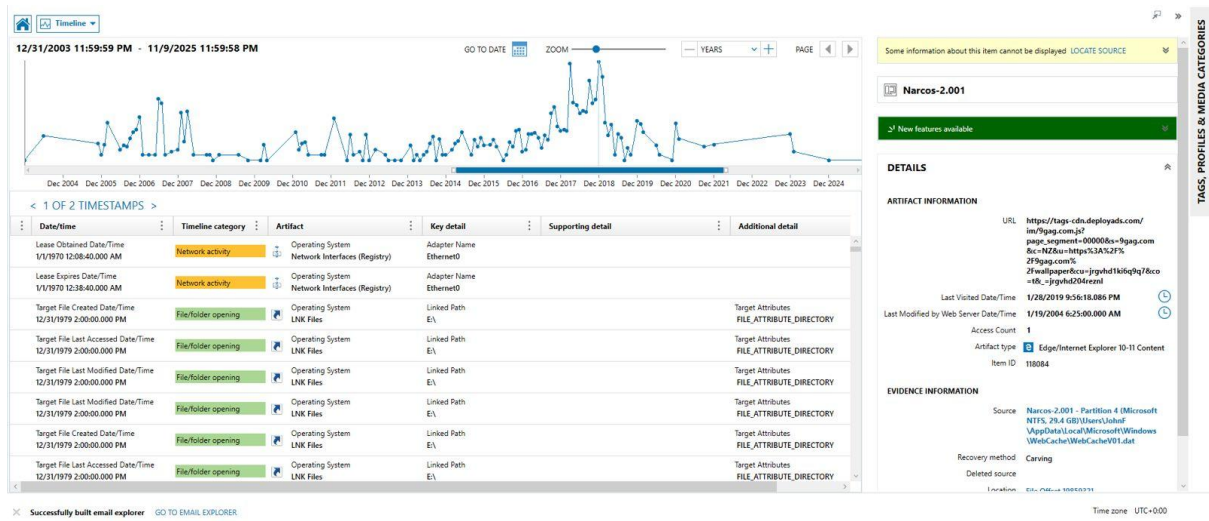


Figure 3: Magnet Axiom Timeline Window

2.1.3. ChromeCacheView

ChromeCacheView was used to extract cached Discord files from local application folders. Since Discord stores messages as cached JSON objects instead of readable logs, this tool was essential for:

- Viewing cached entries
- Exporting cached chat files related to communication between suspects

This tool helped reveal key conversations involving drug discussions, threats, steganography instructions, and meeting arrangements.

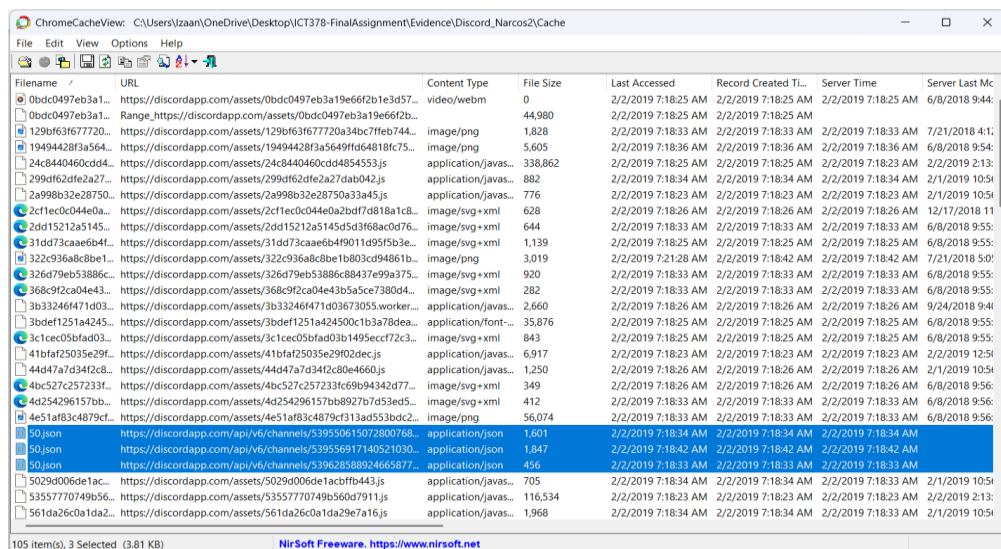


Figure 4: ChromeCacheView Window

2.2. EVIDENCE COLLECTION AND PRESERVATION

Evidence collection followed a strict forensic procedure to ensure the integrity and admissibility of the data. Exact bit-stream copies of the disk images and memory dumps were created using trusted forensic imaging tools. MD5 hash values were calculated for each original dataset and then re-verified after imaging to confirm that identical copies were created. During the investigation, images were mounted in read-only mode to prevent accidental changes. Once the integrity checks were complete, the datasets were processed through Autopsy and AXIOM to extract deleted files, communication logs, browser history, steganography-related content, encrypted containers, and memory artefacts. All findings were documented with timestamps, file paths, and screenshots to ensure clarity, transparency, and a reliable chain of custody throughout the investigation.

3. FINDINGS

3.1. FINDINGS 1 – Identify relevant user account profiles and computer names

associated with the suspect's computers.

3.1.1. DISCUSSION FOR STEVE KOWHAI – INCULPATORY

On Narcos1, the desktop system name and main Windows user profile clearly relate to **Steve**. The username, home folder and system information all point to him as the primary user of the machine seized from 666 Rewera Avenue.

3.1.2. STEVE KOWHAI – SUPPORTING EVIDENCE

Type	Value	Source(s)
Name	SK-DESKTOP	Recent Activity
Program Name	Windows 10 Pro	Recent Activity
Processor Architecture	AMD64	Recent Activity
Temporary Files Directory	%SystemRoot%\TEMP	Recent Activity
Path	C:\Windows	Recent Activity
Product ID	00330-80000-00000-AA502	Recent Activity
Owner	Steve	Recent Activity
Source File Path	/img_Narcos-1.001	
Artifact ID	-9223372036854775642	

Figure 5: Desktop details showing that this was Steve's desktop

Basic Properties	
Login:	steve
Full Name:	
Address:	S-1-5-21-1474204758-2504895174-1356074821-1001
Type:	
Creation Date:	2019-01-28 23:35:06 GST
Object ID:	3463
Narcos-1.001_1 Host Details	
Last Login:	2019-02-02 06:38:22 GST
Login Count:	18
Password Fail Date:	2019-01-30 01:06:35 GST
Password Settings:	Password does not expire, Password not required
Flag:	Normal user account
Home Directory:	C:/Users/Steve

Figure 6: Accounts show Steve's name logged in

3.1.3. DISCUSSION FOR JOHN FREDRICKSON – INCULPATORY

On Narcos2, the main Windows account and laptop name identify John as the primary user. The profile folders, such as C:\Users\[John], and the device naming convention match his identity. This allows the encrypted container, business documents, Discord cache and other artefacts to be reliably attributed to John.

3.1.4. JOHN FREDRICKSON – SUPPORTING EVIDENCE

Type	Value	Source(s)
Name	JOHNFLAPTOP1	Recent Activity
Program Name	Windows 10 Pro	Recent Activity
Processor Architecture	AMD64	Recent Activity
Temporary Files Directory	%SystemRoot%\TEMP	Recent Activity
Path	C:\Windows	Recent Activity
Product ID	00330-80000-00000-AA310	Recent Activity
Owner	JohnF	Recent Activity
Source File Path	/img_Narcos-2.001	
Artifact ID	-9223372036854775569	

Figure 7: Laptop details show that its Johns Laptop

Basic Properties

Login: johnf
 Full Name:
 Address: S-1-5-21-1288840944-4209380227-2421025932-1001
 Type:
 Creation Date: 2019-01-28 23:25:46 GST
 Object ID: 785

Narcos-2.001_1 Host Details

Last Login: 2019-02-02 07:17:41 GST
 Login Count: 12
 Password Fail Date: 2019-02-02 07:17:27 GST
 Password Settings: Password does not expire, Password not required
 Flag: Normal user account
 Home Directory: C:/Users/JohnF

Figure 8: Accounts show Johns account logged in

3.1.5. DISCUSSION FOR JANE ESTEBAN

On Narcos3, the account information links the laptop to Jane. The device is consistent with the “small Windows laptop” mentioned at the airport. While the account mapping itself is neutral, it provides important context so that undercover-related searches and documents can be tied to Jane rather than another user.

3.1.6. JANE ESTEBAN – SUPPORTING EVIDENCE

Type	Value	Source(s)
Name	JELAPTOP	Recent Activity
Program Name	Windows 10 Pro	Recent Activity
Processor Architecture	AMD64	Recent Activity
Temporary Files Directory	%SystemRoot%\TEMP	Recent Activity
Path	C:\Windows	Recent Activity
Product ID	00330-80000-00000-AA263	Recent Activity
Owner	Windows User	Recent Activity
Source File Path	/img_Narcos-3.001	
Artifact ID	-9223372036854775593	

Figure 9: Laptop details show that its Jane's Laptop

Basic Properties

Login: janee
 Full Name:
 Address: S-1-5-21-1418642363-203697023-882285408-1001
 Type:
 Creation Date: 2019-01-28 23:20:32 GST
 Object ID: 4489

Narcos-3.001_1 Host Details

Last Login: 2019-02-02 07:09:36 GST
 Login Count: 13
 Password Hint: gman
 Password Fail Date: 2019-01-30 01:24:18 GST
 Password Settings: Password does not expire, Password not required
 Flag: Normal user account
 Home Directory: C:/Users/JaneE

Figure 10: Accounts show Jane logged in

3.2. FINDING 2 – Identify the roles of each suspect

3.2.1. DISCUSSION FOR STEVE KOWHAI – INCULPATORY

The artefacts on Steve’s desktop show him as the local organizer and receiver in New Zealand. He lives at the backup drop address, researches how to cut crystal meth, stores maps of drop locations, uses privacy-friendly services like ProtonMail, and installs tools such as Image Steganography and CCleaner. He is also the one who first suggests using steganography and encryption to John. These behaviours indicate not only awareness of the criminal activity, but also experience and leadership in handling the technical side of the operation (Rogers et al., 2020).

3.2.2. SUPPORTING EVIDENCE – STEVE KOWHAI

Visit Details

Title: Inbox | crayfish1980@protonmail.com | ProtonMail

Username: Default

Date Accessed: 2019-02-01 04:22:06 GST

Domain: protonmail.com

URL: https://mail.protonmail.com/inbox

Referrer URL: https://mail.protonmail.com/inbox

Program Name: Google Chrome

Source

Host: Narcos-1.001_1 Host

Data Source: Narcos-1.001

File: /img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/History

Figure 11: Shows that he's been using protonmail

History		google.com	best places to trade drugs	Google Chrome	2019-02-02 05:01:34 GST	Narcos-1.001
History		google.com	best places to trade drugs	Google Chrome	2019-02-02 05:01:34 GST	Narcos-1.001
History		google.com	wellington libraries	Google Chrome	2019-02-02 05:01:51 GST	Narcos-1.001
History		google.com	courteney place	Google Chrome	2019-02-02 05:02:51 GST	Narcos-1.001
History		google.com	eastbourne library	Google Chrome	2019-02-02 05:04:36 GST	Narcos-1.001
History		google.com	eastbourne	Google Chrome	2019-02-02 05:05:05 GST	Narcos-1.001
History		google.com	eastbourne library	Google Chrome	2019-02-02 05:05:11 GST	Narcos-1.001

Figure 12: Searches regarding selling drugs, and also searches regarding a location named "eastborn library"

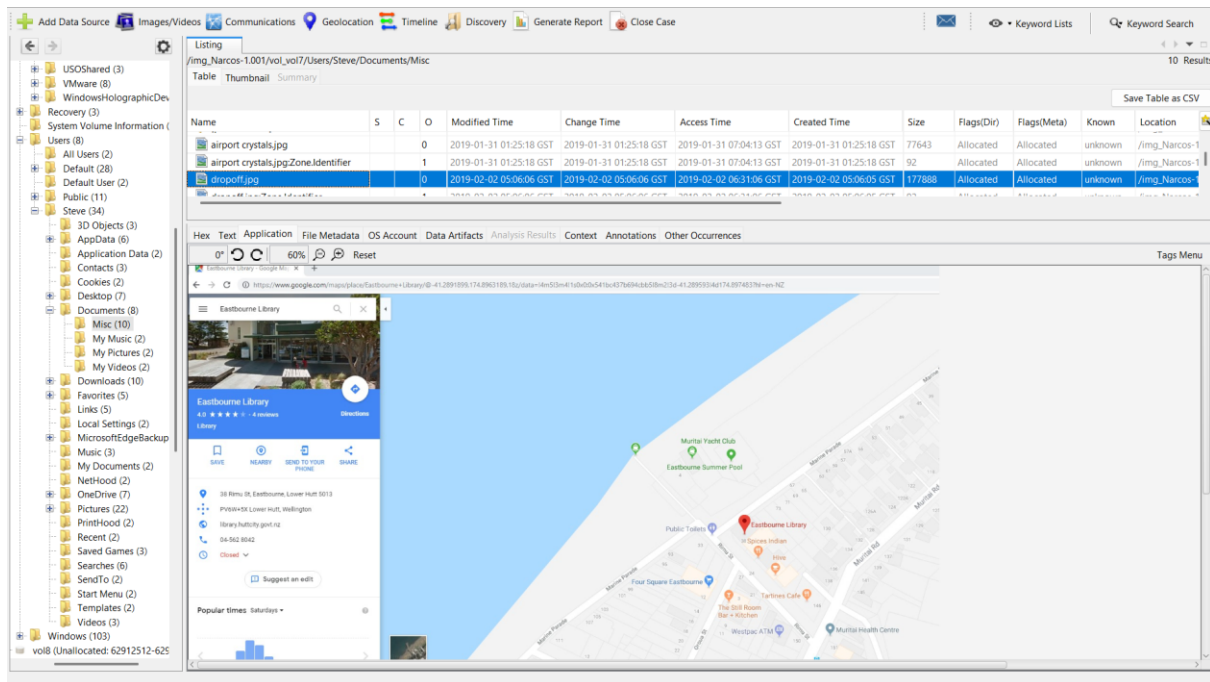


Figure 13: Backup drop address given similar to where he lives

Visit Details	
Title:	Download Image Steganography from SourceForge.net
Username:	Default
Date Accessed:	2019-02-01 04:16:25 GST
Domain:	sourceforge.net
URL:	https://sourceforge.net/projects/image-steg/files/latest/download
Referrer URL:	https://sourceforge.net/projects/image-steg/files/latest/download
Program Name:	Google Chrome
Source	
Host:	Narcos-1.001_1 Host
Data Source:	Narcos-1.001
File:	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/History

Figure 14: Image Steganography Tool download

Metadata	
Name:	/img_Narcos-1.001/vol_vol7/Program Files/CCleaner/CCleaner.exe
Type:	File System
MIME Type:	application/x-dosexec
Size:	14679256
File Name Allocation:	Allocated
Metadata Allocation:	Allocated
Modified:	2019-01-10 14:01:44 GST
Accessed:	2019-02-02 06:38:42 GST
Created:	2019-01-10 14:01:44 GST
Changed:	2019-01-30 03:41:21 GST
MD5:	f0f196fd71b9c7d65929861dd1bcfdcf
SHA-256:	00564fbc89d79ba9aa80a8c2a11c0f721e5ffda0ca0549273a22cf270e52664f
Hash Lookup Results:	UNKNOWN
Internal ID:	30207

Figure 15: Running CCleaner

3.2.3. DISCUSSION FOR JOHN FREDRICKSON – INCULPATORY

John appears to be the central coordinator. His laptop holds organized client and shipment files, flight planning material and Discord communication with multiple parties. He gives instructions, arranges travel and negotiates with Steve and other contacts. The structure of his documents and his dominant tone in chats support the view that he managed the business side of the smuggling scheme.

3.2.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

	A	B	C	D	E
1	Name	Location	Product	Amount	Delivery
2	Ricky Ross	Los Angeles	Mama Coca	20kg	Monthly
3	Frank Lucas	New York, USA	Ferry Dust	15kg	Quarterly
4	Chris Coke	Kingston Jamaica	Coke	20kg	Monthly
5	Steve Kowhai	Wellington, New Zealand	Crank	15kg	Monthly
6	Don Cholito	Puerto Rico	Snow	25kg	Quarterly
7	Manuel Noriega	Panama	Smack	15kg	Monthly
8	Joaquin Guzman	Guadalajara, Mexico	China White	15kg	Monthly
9	Leroy Barnes	New York, USA	Load pack	15kg	Quarterly
10	AL Capone	Sicily, Italy	Silly putty	25kg	Monthly
11	Jane Esteban	Brisbane, Australia	Uppers	1 gram	On demand
12	Pablo Escobar	Colombia	White horse	15kg	Quarterly
13	Franz Sanchez	Isthmus City	Mary Jane	20kg	Quarterly
14	Jake Heke	Auckland	Tweak	10kg	Monthly
15					

Figure 16: A file called “clients” found in “business” folder

Track this shipment via the DHL Web Site : <http://www.dhl.com>

Shipment Air Waybill
(Non negotiable)

ORIGIN B N E **DESTINATION CODE** A K L

1 Payer account number and insurance details
Charge to ☒ Shipper ☐ Receiver ☐ 3rd party ☒ Cash ☐ Cheque ☐ Credit Card
Payer Account No. 001-158545-85
Shipment Insurance see reverse
☒ Yes Insured value (in local currency) 0 Not all payment options are available in all countries.

2 From (Shipper)
Shipper's account number 258-85695 Contact name Johnny Fredrick
Shipper's reference (up to 32 characters but only first 12 will be shown on invoice) AB-20071223-589X
Company name High As a Kite LLC
Address 8515 Haven Wood Trail
Inala, Brisbane
QLD 4077
Australia
Postcode/Zip Code (required) QLD 4077 Phone, Fax or E-mail (required) +1 258 585 965

3 To (Receiver)
Company name
Delivery address DHL cannot deliver to a PO Box
5/34 Hapua Street
Remuera
Auckland 1050
New Zealand
Postcode/Zip Code (required) 1050 Country New Zealand
Contact person Jake Heke Phone, Fax or E-mail (required) +6402145365477

4 Shipment details
Total number of packages 1 Total Weight 20kg
Dimensions in cm
Length Width Height
500mm x 500mm x 600mm

5 Full description of contents
Give content and quantity
1x Pressure cooker
3x Pots
1x Bread Maker

6 Non-Document Shipments Only (Customs Requirement)
Attach the original and four copies of a Proforma or Commercial invoice
Shipper's VAT/GST number Receiver's VAT/GST or Shipper's EIN/SSN
Declared Value for Customs (as on commercial/proforma invoice) Harmonised Commodity Code if applicable
TYPE OF EXPORT ☒ Permanent ☐ Repair / Return ☐ Temporary
Destination duties/taxes if left blank receiver pays duties/taxes
☒ Receiver ☐ Shipper ☐ Other specify approved account number

7 Shipper's agreement (Signature required)
Unless otherwise agreed in writing, I/we agree that DHL's Terms and Conditions of Carriage are all the terms of the contract between me/us and DHL and (1) such Terms and Conditions and, where applicable, the Warsaw Convention limits and/or excludes DHL's liability for loss, damage or delay and (2) this shipment does not contain cash or dangerous goods (see reverse).
Signature Johnny Fredrick Date 29 / 01 / 2019

8 Services
☐ Domestic ☐ International ☐ International Document ☐ International Non Document ☐ Europe
Not all Services are available to and from all locations
☐ Express 9 (10.30 to the USA)
☐ Express 12
☐ Express / Worldwide
☐ Express Envelope
☐ Other
Optional Services (extra charges may apply)
☐ Saturday Delivery ☐ Special Pick-Up
☐ Delivery Notification
☐ Other
DHL Global Mail ☐ GMB Priority ☐ GMB Standard ☐ Other

9 DIMENSIONAL/CHARGEABLE WEIGHT
kg • gr
CHARGES Services
Other
Insurance
VAT
CURRENCY TOTAL
TRANSPORT COLLECT STICKER No.
PAYMENT DETAILS (Cheque, Card No.)
No. :
Type Expires
Picked up by
Route No.
Time Date

PARAGON - BLUE201007DR - 121750 VERSION 1 3rd proof

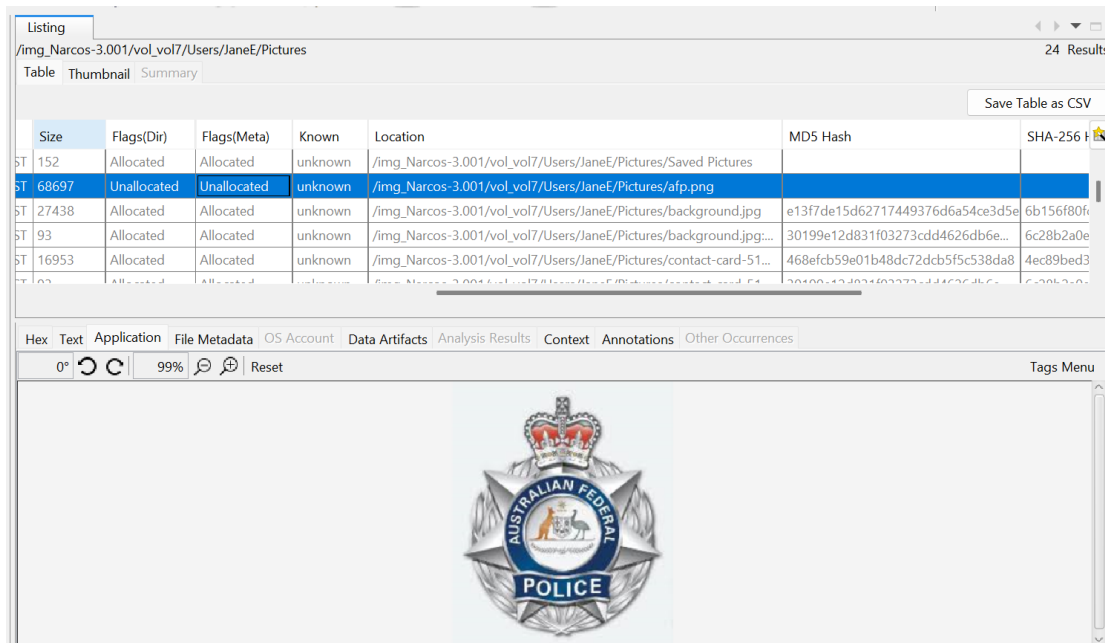
Origin copy

Figure 17: Shipment details

3.2.5. DISCUSSION FOR JANE ESTEBAN – EXCULPATORY

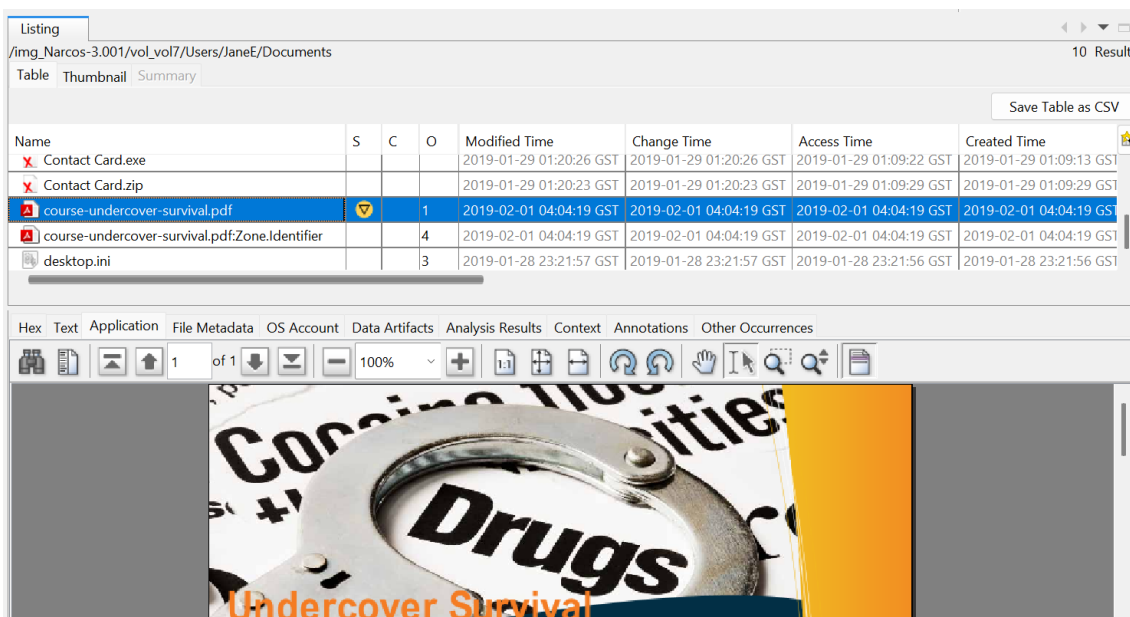
Jane's artefacts tell a different story. Her device contains multiple pictures proving that she's an undercover cop, also undercover survival guide document was found in her laptop, practice material about how to act like an addict. Together, these items support the conclusion that she is an undercover AFP officer whose role was to gather evidence.

3.2.6. SUPPORTING EVIDENCE – JANE ESTEBAN



	Size	Flags(Dir)	Flags(Meta)	Known	Location	MDS Hash	SHA-256 Hash
ST 152		Allocated	Allocated	unknown	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/Saved Pictures		
ST 68697		Unallocated	Unallocated	unknown	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/afp.png		
ST 27438		Allocated	Allocated	unknown	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/background.jpg	e13f7de15d62717449376d6a54ce3d5e	6b156f80f
ST 93		Allocated	Allocated	unknown	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/background.jpg...	30199e12d831f03273cdd4626db6e...	6c28b2a0e
ST 16953		Allocated	Allocated	unknown	/img_Narcos-3.001/vol_vol7/Users/JaneE/Pictures/contact-card-51...	468efcb59e01b48dc72dc5f5c538da8	4ec89bed3

Figure 18: she was an undercover agent



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
Contact Card.exe				2019-01-29 01:20:26 GST	2019-01-29 01:20:26 GST	2019-01-29 01:09:22 GST	2019-01-29 01:09:13 GST
Contact Card.zip				2019-01-29 01:20:23 GST	2019-01-29 01:20:23 GST	2019-01-29 01:09:29 GST	2019-01-29 01:09:29 GST
course-undercover-survival.pdf			1	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST
course-undercover-survival.pdf:Zone.Identifier			4	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST	2019-02-01 04:04:19 GST
desktop.ini			3	2019-01-28 23:21:57 GST	2019-01-28 23:21:57 GST	2019-01-28 23:21:56 GST	2019-01-28 23:21:56 GST

Figure 19: Evidence indicating she wanted to study the measures to be taken to survive as an undercover cop

3.3. FINDINGS 3 – Identify relevant web activity on the suspect's computer.

3.3.1. DISCUSSION FOR STEVE KOWHAI – INCULPATORY

Steve's web history and RAM artefacts reveal repeated searches on crystal meth, including how to cut or prepare it and related drug information. There are also visits to mapping websites used to check addresses and drug routes into Wellington. These activities show that Steve was researching drug handling in advance and was actively looking up information for the delivery.

3.3.2. SUPPORTING EVIDENCE – STEVE KOWHAI

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				google.com	google maps	Google Chrome	2019-01-31 01:17:23 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:07 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:07 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:24 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:24 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:24 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:30 GST	Narcos-1.001
History				google.com	crystal meth	Google Chrome	2019-01-31 06:56:33 GST	Narcos-1.001
History				google.com	drug paraphernalia	Google Chrome	2019-01-31 06:57:16 GST	Narcos-1.001
History				google.com	drug paraphernalia	Google Chrome	2019-01-31 06:57:21 GST	Narcos-1.001
History				google.com	drug paraphernalia	Google Chrome	2019-01-31 06:57:21 GST	Narcos-1.001
WebCacheV01.dat				bing.com	crayfish diving nz	Microsoft Edge Analyzer	2019-01-28 22:01:12 GST	Narcos-1.001
WebCacheV01.dat				google.com	cutting drugs	Microsoft Edge Analyzer	2019-01-28 22:58:52 GST	Narcos-1.001
WebCacheV01.dat				google.com	cutting drugs	Microsoft Edge Analyzer	2019-01-28 22:59:54 GST	Narcos-1.001
WebCacheV01.dat				google.com	cutting drugs	Microsoft Edge Analyzer	2019-01-28 23:00:21 GST	Narcos-1.001
WebCacheV01.dat				google.com	cutting drugs	Microsoft Edge Analyzer	2019-01-28 23:00:22 GST	Narcos-1.001
WebCacheV01.dat				google.com	cutting agents for ice	Microsoft Edge Analyzer	2019-01-28 23:02:00 GST	Narcos-1.001
WebCacheV01.dat				google.com	how to launder money	Microsoft Edge Analyzer	2019-01-28 23:03:51 GST	Narcos-1.001
WebCacheV01.dat				bing.com	drug routes in wellington	Microsoft Edge Analyzer	2019-01-29 01:01:58 GST	Narcos-1.001
WebCacheV01.dat				bing.com	google.co.nz	Microsoft Edge Analyzer	2019-01-29 01:02:12 GST	Narcos-1.001
WebCacheV01.dat				google.co.nz	drug routes in wellington	Microsoft Edge Analyzer	2019-01-29 01:02:39 GST	Narcos-1.001
WebCacheV01.dat				google.co.nz	drug routes in wellington	Microsoft Edge Analyzer	2019-01-29 01:02:52 GST	Narcos-1.001
WebCacheV01.dat				google.co.nz	drug routes in around wellington	Microsoft Edge Analyzer	2019-01-29 01:03:42 GST	Narcos-1.001
WebCacheV01.dat				google.co.nz	drug routes in around wellington	Microsoft Edge Analyzer	2019-01-29 01:03:47 GST	Narcos-1.001

Figure 20: Steve's Relevant Web Activity

3.3.3. DISCUSSION FOR JOHN FREDRICKSON

John's browsing includes searches and visits related to flights from Brisbane to Wellington, booking pages, as well as downloads and information for tools such as Image Steganography, indicating that he was trying to hide information. His web activity shows him to trying to find ways to smuggle drugs and knowing about drug smuggling.

3.3.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

Visit Details

Title: Last Minute Brisbane to Wellington Flights - Cheap Airfares
 Date Accessed: 2019-01-31 06:16:56 GST
 Domain: lastminute.com.au
 URL: https://www.lastminute.com.au/vc/flights/brisbane-bne-to-wellington-wlg/
 Program Name: FireFox Analyzer

Figure 21: Web search shows Johns intention to travel to Wellington

The screenshot shows a web browser history table with columns: Source Name, S, C, O, URL, Date Accessed, Referrer URL, Title, and Program. The table lists several history entries, including Google searches for 'image steganography download' and 'protonmail', and ProtonMail login attempts. The selected entry is a Google search for 'image steganography download'.

Source Name	S	C	O	URL	Date Accessed	Referrer URL	Title	Program
History		4		https://www.google.com/search?q=all+black&oeq=all...	2019-02-01 01:32:55 GST	https://www.google.com/search?q=all+black&oeq=all...	all blacks - Google Search	Google
History		1		https://www.stuff.co.nz/sport/rugby/super-rugby/11027...	2019-02-01 01:32:59 GST	https://www.stuff.co.nz/sport/rugby/super-rugby/11027...	All Blacks and Crusaders hooker Codie Taylor to take ex...	Google
History		4		https://www.google.com/search?q=protonmail&oeq=pr...	2019-02-01 04:12:20 GST	https://www.google.com/search?q=protonmail&oeq=pr...	protonmail - Google Search	Google
History		4		https://protonmail.com/	2019-02-01 04:12:23 GST	https://protonmail.com/	Secure email: ProtonMail is free encrypted email.	Google
History		4		https://mail.protonmail.com/login	2019-02-01 06:40:19 GST	https://mail.protonmail.com/login	Login - ProtonMail	Google
History		4		https://mail.protonmail.com/login/unlock	2019-02-01 04:22:02 GST	https://mail.protonmail.com/login/unlock	Login ProtonMail	Google
History		4		https://mail.protonmail.com/inbox	2019-02-01 04:22:06 GST	https://mail.protonmail.com/inbox	Inbox crayfish1980@protonmail.com ProtonMail	Google
History		4		https://mail.protonmail.com/inbox/OMi2pw6297Pp106...	2019-02-01 04:21:52 GST	https://mail.protonmail.com/inbox/OMi2pw6297Pp106...	Login - ProtonMail	Google
History		4		https://www.google.com/search?q=image+steganogra...	2019-02-01 04:15:04 GST	https://www.google.com/search?q=image+steganogra...	image steganography download - Google Search	Google
History		2		https://sourceforge.net/projects/image-steg/	2019-02-01 06:39:56 GST	https://sourceforge.net/projects/image-steg/	Image Steganography download Sourceforge.net	Google

Below the table, the 'Visit Details' for the selected entry are shown:

Visit Details
 Title: image steganography download - Google Search
 Username: Default
 Date Accessed: 2019-02-01 04:15:04 GST
 Domain: google.com
 URL: https://www.google.com/search?q=image+steganography+download&oeq=image+steganography+down&aeq=chrome.0.0.69.57.0.4.40639.0.8&sourceid=chrome&ie=UTF-8
 Referrer URL: https://www.google.com/search?q=image+steganography+download&oeq=image+steganography+down&aeq=chrome.0.0.69.57.0.4.40639.0.8&sourceid=chrome&ie=UTF-8
 Program Name: Google Chrome

Source
 Host: Narcos-1.001_1 Host
 Data Source: Narcos-1.001
 File: /img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/History

Figure 22: Researching about image steganography and downloading a tool for it

Visit Details

Title: FYI: How Much Cocaine Can You Fit In Your, Ahem, Body? | Popular Science
 Date Accessed: 2019-01-31 06:10:26 GST
 Domain: popsci.com
 URL: https://www.popsci.com/science/article/2013-04/how-much-cocaine-can-you-fit-your-body#page-2
 Referrer URL: https://www.popsci.com/science/article/2013-04/how-much-cocaine-can-you-fit-your-body
 Program Name: FireFox Analyzer

Source

Host: Narcos-2.001_1 Host
 Data Source: Narcos-2.001
 File: /img_Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/fj248tbu.default/places.sqlite

Figure 23: Web searches regarding smuggling drugs

3.3.5. DISCUSSION FOR JANE ESTEBAN

Jane's web activity is centred around undercover behaviour and survival, not around selling or distributing drugs. She searches for how to look like a meth addict, how to maintain an undercover role. This supports her status as an officer in disguise rather than a willing drug trafficker.

3.3.6. SUPPORTING EVIDENCE – JANE ESTEBAN

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
WebCacheV01.dat				google.com.au	federal agent badge	Microsoft Edge Analyzer	2019-01-28 22:41:58 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	federal agent badge au	Microsoft Edge Analyzer	2019-01-28 22:42:12 GST	Narcos-3.001
WebCacheV01.dat				bing.com	disocrd	Microsoft Edge Analyzer	2019-01-29 00:01:29 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	micro voice recorprder	Microsoft Edge Analyzer	2019-01-29 00:05:25 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	micro voice recorder au	Microsoft Edge Analyzer	2019-01-29 00:05:37 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how to act like a meth addict	Microsoft Edge Analyzer	2019-01-29 00:08:21 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how to look like a meth addict	Microsoft Edge Analyzer	2019-01-29 00:10:50 GST	Narcos-3.001
WebCacheV01.dat				bing.com	news9	Microsoft Edge Analyzer	2019-01-29 21:27:42 GST	Narcos-3.001
WebCacheV01.dat				bing.com	theguardian	Microsoft Edge Analyzer	2019-01-29 21:29:49 GST	Narcos-3.001
WebCacheV01.dat				bing.com	reddit	Microsoft Edge Analyzer	2019-01-29 21:31:04 GST	Narcos-3.001

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
WebCacheV01.dat				bing.com	redditr	Microsoft Edge Analyzer	2019-01-30 21:35:40 GST	Narcos-3.001
WebCacheV01.dat				bing.com	prison sentences for drugs	Microsoft Edge Analyzer	2019-01-31 01:05:27 GST	Narcos-3.001
WebCacheV01.dat				bing.com	prison sentences for drugs au	Microsoft Edge Analyzer	2019-01-31 01:05:41 GST	Narcos-3.001
WebCacheV01.dat				bing.com	prison sentences for drugs nz	Microsoft Edge Analyzer	2019-01-31 01:06:04 GST	Narcos-3.001
WebCacheV01.dat				bing.com	passing drugs through nz customs	Microsoft Edge Analyzer	2019-01-31 01:06:39 GST	Narcos-3.001
WebCacheV01.dat				bing.com	google	Microsoft Edge Analyzer	2019-01-31 01:08:20 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	methamphetamine	Microsoft Edge Analyzer	2019-01-31 01:08:35 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	methamphetamine crystal rock	Microsoft Edge Analyzer	2019-01-31 01:08:42 GST	Narcos-3.001
WebCacheV01.dat				google.com.au	methamphetamine crystal rock	Microsoft Edge Analyzer	2019-01-31 01:08:44 GST	Narcos-3.001
WebCacheV01.dat				bing.com	ch9 news	Microsoft Edge Analyzer	2019-01-31 21:30:58 GST	Narcos-3.001

Figure 24: Jane's Web Searches

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
WebCacheV01.dat				google.com.au	methamphetamine crystal rock	Microsoft Edge Analyzer	2019-01-31 01:08:44 GST	Narcos-3.001
WebCacheV01.dat				bing.com	ch9 news	Microsoft Edge Analyzer	2019-01-31 21:30:58 GST	Narcos-3.001
WebCacheV01.dat				bing.com	legal process to convict blackmail	Microsoft Edge Analyzer	2019-01-31 22:59:56 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how tp pretend to be desperate	Microsoft Edge Analyzer	2019-01-31 23:05:45 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how to pretend to be desperate	Microsoft Edge Analyzer	2019-01-31 23:05:47 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how to pretend to be desperate in drug dealings	Microsoft Edge Analyzer	2019-01-31 23:06:33 GST	Narcos-3.001
WebCacheV01.dat				bing.com	how to pretend to be desperate	Microsoft Edge Analyzer	2019-01-31 23:07:16 GST	Narcos-3.001
WebCacheV01.dat				bing.com	survival tips from an undercover cop	Microsoft Edge Analyzer	2019-01-31 23:55:31 GST	Narcos-3.001
WebCacheV01.dat				bing.com	survival tips from an undercover cop	Microsoft Edge Analyzer	2019-02-01 00:01:50 GST	Narcos-3.001
WebCacheV01.dat				bing.com	undercover cop survival	Microsoft Edge Analyzer	2019-02-01 00:03:11 GST	Narcos-3.001

Figure 25: Searching for tips to survive as an undercover cop

3.4. FINDINGS 4 – Identify images that help to build a profile of the suspect's behavior

3.4.1. DISCUSSION FOR STEVE KOWHAI – INCULPATORY

Steve's desktop contains several deleted images that appear to show drug-related content. At the same time, there are screenshots of online maps and route diagrams related to delivery points. The fact that some pictures are deleted and had to be recovered supports the idea that Steve consciously tried to remove incriminating visual evidence, and these images help build a profile of him as someone comfortable around drugs and organised crime.

3.4.2. SUPPORTING EVIDENCE – STEVE KOWHAI



Figure 26: Pictures of potential street gang



Figure 27: Other Pictures found in Recycle Bin

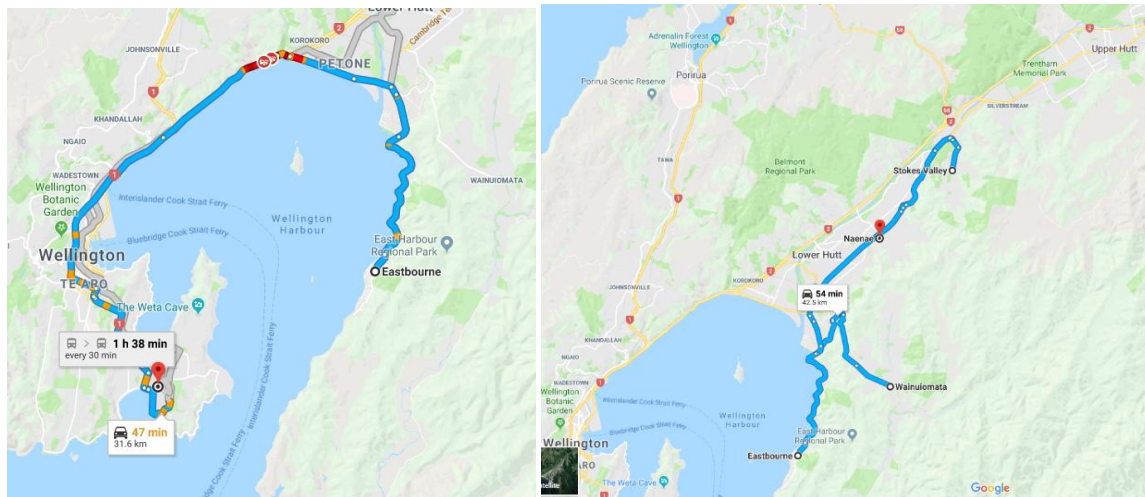


Figure 28: Map showing directions to a certain area

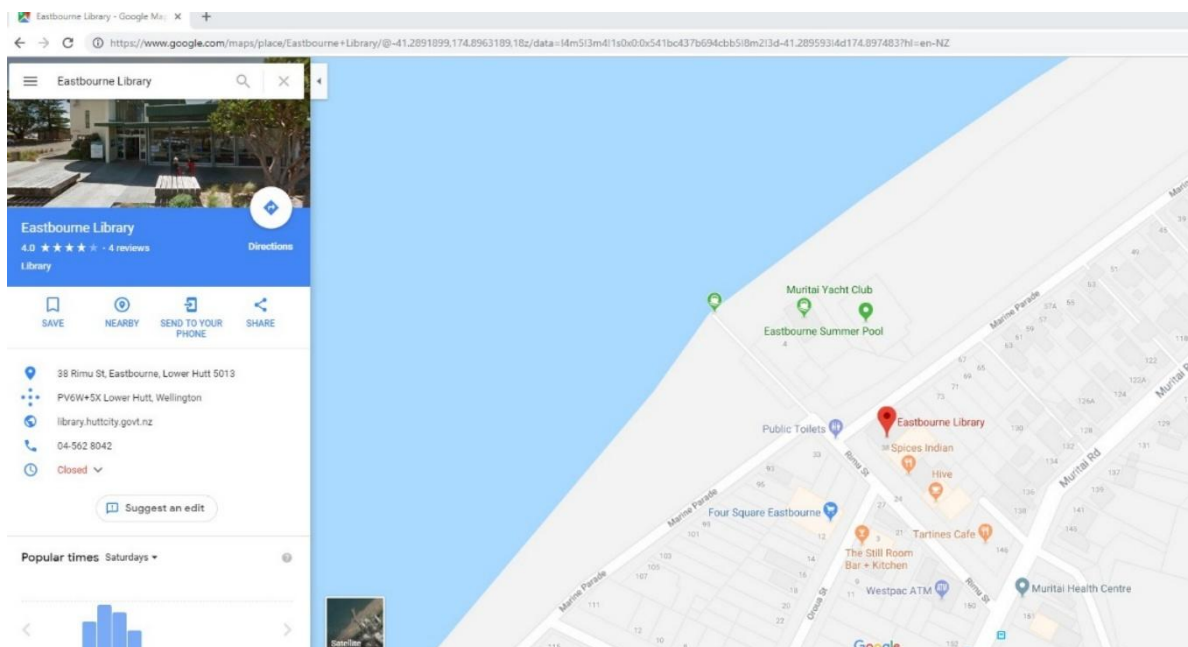


Figure 29: Another screenshot of a map saved as "drop off" showing a potential delivery address

3.4.3. DISCUSSION FOR JOHN FREDRICKSON – INCULPATORY

Images on John's laptop include flight screenshots, drugs smuggling methods and other images extracted from steganographic containers such as "package.jpg". These pictures link strongly to planning and concealment of the smuggled goods. The combination of travel images and embedded images hidden inside other files demonstrates his insight into both logistics and covert communication.

3.4.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

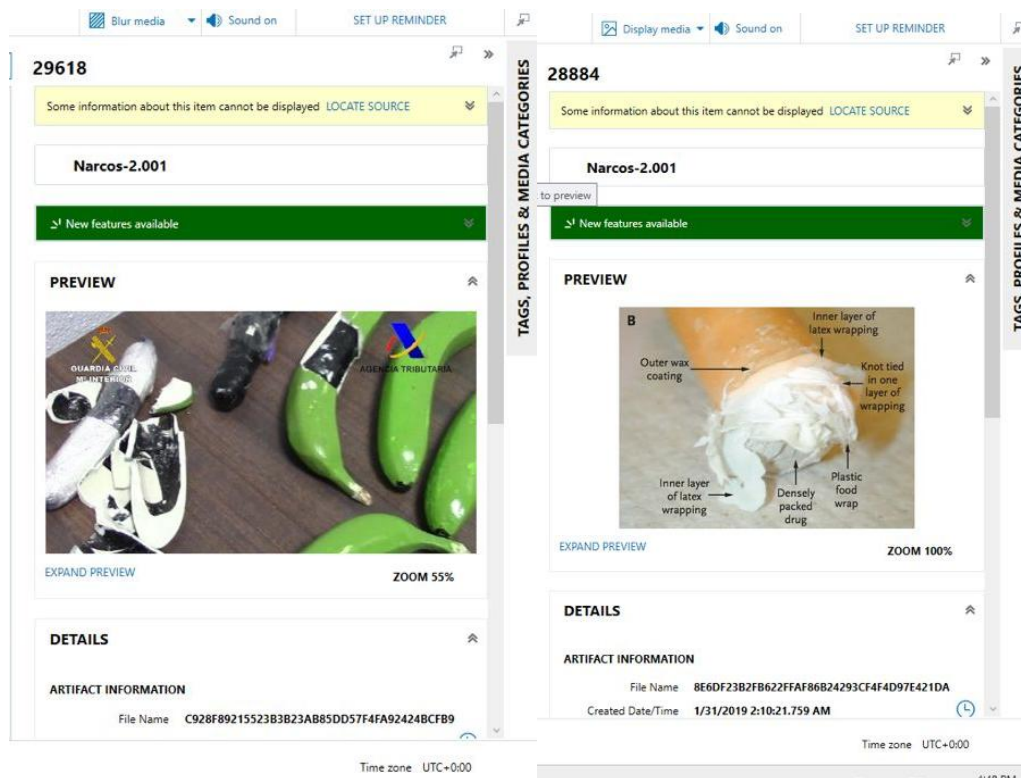


Figure 30: images found related to smuggling drugs



Figure 31: Hidden file "package.jpg" extracted using a decrypting tool

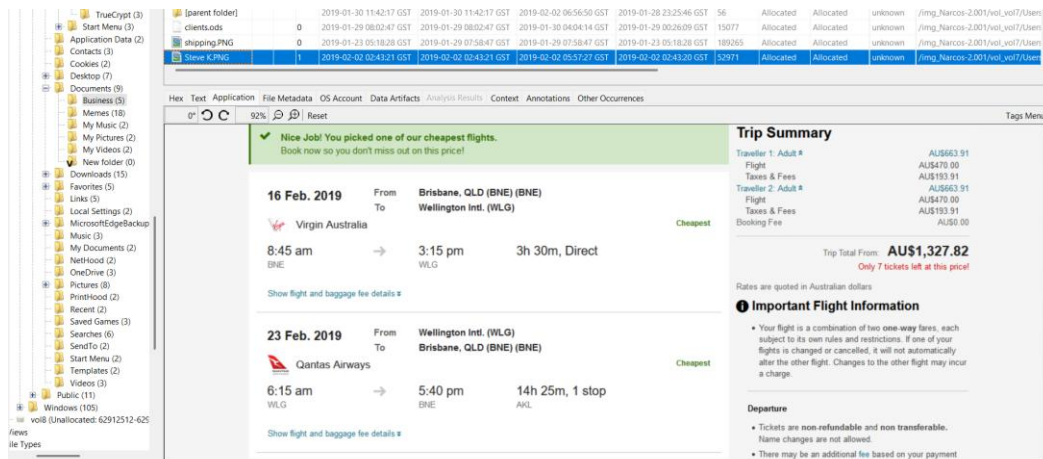


Figure 32: Flight screenshot

3.4.5. DISCUSSION FOR JANE ESTEBAN – EXCULPATORY

Jane's images are mostly personal confirming her position as a cop, with no strong indication that she was preparing or handling drugs. Some sensitive images, such as those used to threaten her about her "children", appear inside conversations rather than as original files created by her. These images are important in understanding the pressure she was under, but they do not show her planning or benefiting from drug activity.

3.4.6. SUPPORTING EVIDENCE – JANE ESTEBAN



Figure 33: Pictures showing her position



Figure 34: Jane's children's pics sent to her threatening her

3.5. FINDINGS 5 – Identify binary files that could help the investigation.

3.5.1. DISCUSSION FOR STEVE KOWHAI – INCULPATORY

On Steve’s system, the presence of Image Steganography software and wiping tools such as CCleaner is significant. These binaries show that Steve had the software needed to hide data in images and to clear traces of his activity after use. Even if not every run can be tied to a specific offence, their installation supports the idea that he actively maintained a toolkit for covert communication and system cleaning.

3.5.2. SUPPORTING EVIDENCE – STEVE KOWHAI

	Apache_OpenOffice_4.1.6_Win_x86_install_en-US.exe		1	2019-01-29 00:43:32 GST
	ccsetup552.exe		0	2019-01-30 03:41:07 GST
	ccsetup552.exe:Zone.Identifier		0	2019-01-30 03:41:07 GST
	ChromeSetup.exe		1	2019-01-30 01:05:06 GST
	desktop.ini		3	2019-01-28 23:35:35 GST
	Image Steganography 1.5.2 Setup.exe		1	2019-02-01 04:16:34 GST
	TrueCrypt Setup 7.1a.exe		2	2019-01-30 03:27:51 GST

Figure 35: Some binary files found in Steves's Desktop

3.5.3. DISCUSSION FOR JOHN FREDRICKSON – INCULPATORY

Key binary files on John’s laptop include the TrueCrypt installer, the encrypted container itself, and the Quasar RAT bundle he executed. These binaries demonstrate that he set up encryption, opened an encrypted container containing a virus, ran that which allowed remote access to his system.

3.5.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

	Apache_OpenOffice_4.1.6_Win_x86_install_en-US.exe		1	2019-01-29 00:21:27 GST
	Attachments-Important, crucial to our method.zip		0	2019-01-30 04:00:16 GST
	Attachments-Important, crucial to our method.zip:Zone.Identifier		0	2019-01-30 04:00:16 GST
	BavPro_Setup_Mini_C1.exe		0	2019-01-29 00:17:25 GST
	Contact_Card.zip		2	2019-01-30 02:01:31 GST
	Contact_Card.zip:Zone.Identifier		0	2019-01-30 02:01:31 GST
	desktop.ini		3	2019-01-30 11:42:16 GST
	DiscordSetup.exe		1	2019-01-29 00:21:04 GST
	DiscordSetup.exe:Zone.Identifier		2	2019-01-29 00:21:04 GST
	Image Steganography 1.5.2 Setup.exe		1	2019-02-01 03:00:17 GST
	TrueCrypt Setup 7.1a.exe		2	2019-01-30 03:58:55 GST

Figure 36: binary files found in John's laptop – includes “contact_card” which contained a Quasar RAT virus, sent by Jane to gain access to his device

3.5.5. DISCUSSION FOR JANE ESTEBAN – EXCULPATORY / MIXED

The main binary of interest on Jane’s device is the Quasar package sent to John. In another context this would be clearly malicious, but given the surrounding undercover documentation and case details, it appears to have been used as a law-enforcement tool.

3.5.6. SUPPORTING EVIDENCE – JANE ESTEBAN

	Contact Card.zip		2	2019-01-29 07:32:31 GST
	contact_card_512_CA3_icon.ico		2	2019-01-29 01:23:47 GST
	contact_card_512_CA3_icon.ico:Zone.Identifier		2	2019-01-29 01:23:47 GST

Figure 37: Quasar Package named as “contact card” Jane assembled to send to John

3.6. FINDINGS 6 – Identify the means and the content of communications between all the suspects.

3.6.1. DISCUSSION FOR STEVE KOWHAI & JOHN FREDRICKSON– INCULPATORY

Steve and John communicates mainly through Discord and secure email. Discord chat logs show them discussing drug “samples”, delivery timing, and the exchange of image-based messages. The use of Discord and ProtonMail suggests they prefer channels that are harder to monitor. John is the centre of the communication network. His messages often show control, urgency and frustration, which is consistent with his role as organiser.

3.6.2. SUPPORTING EVIDENCE – STEVE KOWHAI & JOHN FREDRICKSON

	O	Q	T
1		author	
2	username	id	content
3	crayfish1980	539549564202516481	Good. Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone.
4	heresjohnny1	539540677894078467	See you soon. John out
5	crayfish1980	539549564202516481	Ya.. I just told you about the tool  Received it. Will check to see if it works and confirm soon.
6	heresjohnny1	539540677894078467	It worked , sending it and the password via email now. I used a tool called image steganography.
7	heresjohnny1	539540677894078467	Okay, I'll have a look and see if I can get it to work and then send the image through
8	crayfish1980	539549564202516481	A way of hiding one image within another. There's a simple application called 'Image Steganography'.
9	heresjohnny1	539540677894078467	No what's that?
10	crayfish1980	539549564202516481	Good Thinking, I already know how. Heard of steganography?
11	heresjohnny1	539540677894078467	I want to send an image an image of something to you but it needs to be done safely. Any ideas?
12	heresjohnny1	539540677894078467	Great. John Out.
13	crayfish1980	539549564202516481	Yeah yeah probably wiser, good one. In fact I have already put a document together in anticipation of chatting with you. I'll send it through now.
14			It contains some information regarding how I see this working out. Let me know what you think once you have read it. S Out.
15	heresjohnny1	539540677894078467	Hmm 10 is a bit much for the first time around and is pretty risky. How about we start off with 1 and can ramp up from there if all goes smoothly?
16	crayfish1980	539549564202516481	New supplier eh? Definitely Interested! Can I get 10 keys of it delivered to Wellington
17	heresjohnny1	539540677894078467	Got a new supplier. Ya Interested??
18	heresjohnny1	539540677894078467	Okay. Talk tomorrow, gotta run
19	crayfish1980	539549564202516481	Need to get a new delivery
20	heresjohnny1	539540677894078467	
21			

Figure 38: Discords chats between Steve and John (screenshot shows chats in down to up form)

	I	J	K	L M	N
1		author		r	
2	username	discrimin id		a e	content
3	jake.heke89	9150 539624169587998720		Ok sweet 10kg is good	5
4	heresjohnny1	5340 539540677894078467		Planning a shipment within the next month. Shut up and wait. How much do you want?	5
5	jake.heke89	9150 539624169587998720		Yo, need the shipment ASAP	5
6	jake.heke89	9150 539624169587998720			5

Figure 39: Another chat between john and jake(customer) showing john as the one dealing with clients

3.6.3. DISCUSSION FOR JANE ESTEBAN

Jane's communications clearly show a power imbalance. Her messages to John reveal fear, and John's replies include threats and blackmail. She gives status updates and confirms instructions in a way that suggests compliance rather than leadership. These communications strongly support the argument that she was acting under duress, not as a willing partner.

3.6.4. SUPPORTING EVIDENCE – JANE ESTEBAN

1	author		
2	username		content
3	becomingjane	I'll be there	
4	heresjohnny1	Flights booked. I'll pick you up from the Woolworths (133 Oxley Station Rd, Oxley QLD 407, Australia) at ... Just bring yourself I'll cover everything else	
5	becomingjane	My full name is Jane Esteban and my birthday is 13/07/1992	
6	heresjohnny1	Right. What's your full name and date of birth? I need it for booking the flights ASAP.	
7	heresjohnny1	Good. Talk tomorrow at 3PM or else	
8	becomingjane	Yes John... I got it	
9	heresjohnny1	Good, now that's what I wanted to hear. Here is the plan. You and I will be acting like a couple travelling on a holiday to New Zealand.	
10		This is what I want you to do: Look the part, act normal, and don't tell anyone about what we're doing. Understood?	
11	becomingjane	WHAT! Please, I swear whatever you need I'll do it... I've put them through enough as it is. What do you want from me??	
12	heresjohnny1	You don't wanna see them hurt do ya	
13	heresjohnny1	I didn't want to do this but, you leave me no choice	
14	becomingjane	Err nah I'm not keen	
15	heresjohnny1	Yea I understand but ill make it worth your while	
16	becomingjane	Umm I don't know, sounds pretty risky.	
17	heresjohnny1	I have a business dealing happening overseas and I need you to accompany me as cover. Everything will be paid for but you need to keep your trap shut	
18	becomingjane	What is it... ?	
19	heresjohnny1	Aight cool, umm actually I have a proposition for ya	
20	becomingjane	The usual	
21	heresjohnny1	Also that contact card didn't open and I can't be bothered with new clients right now... Bigger fish to fry	
22	heresjohnny1	Yo, what you after?	
23	heresjohnny1	Sweet thanks	
24	becomingjane	Also I've got some friends that want to score too. Here's their contact card	
25	becomingjane	Got any of that ice??	

Figure 40: Discord chats between Jane and John (down to up form)

3.7. FINDINGS 7 – Identify any documents that could help the investigation

3.7.1. DISCUSSION FOR STEVE KOWHAI

Steve’s device has fewer user-created documents, but documents like “user guide” for certain software’s like TrueCrypt were found, these type of documents help confirm his involvement.

3.7.2. SUPPORTING EVIDENCE – STEVE KOWHAI

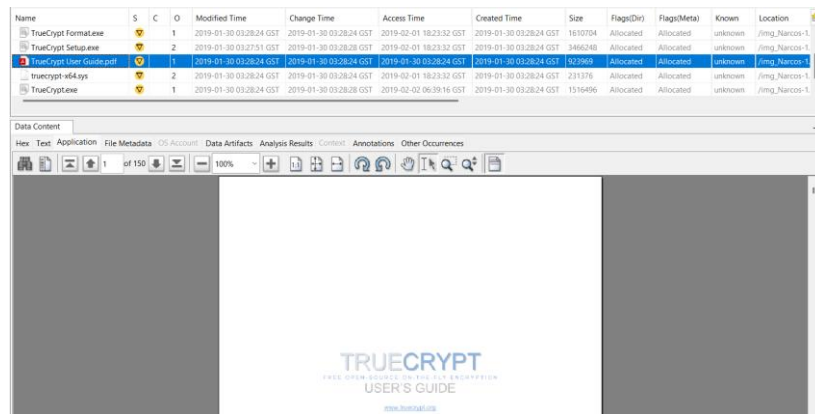


Figure 41: User Guide found for TrueCrypt

3.7.3. DISCUSSION FOR JOHN FREDRICKSON

John’s documents are central to the investigation. The smuggling memo hidden inside the secret file outlines methods and practical considerations for transporting drugs. Additional documents such as client lists and shipment records show that he tracks people and quantities in a structured way. These files strongly indicate deliberate planning and repeated involvement.

3.7.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

Crystal Method

Destination – Wellington
Product – 1 kg crystal
Source – Brisbane (JF)

Wellington is part of the lower north island of New Zealand. It is accessible by road and sea. As tempting as the sea is, be aware that the coast of NZ may seem ideal, but everyone knows it's vulnerable as there have been several news reports about it.

The product is coming from Brisbane, Australia and as a result, will need to be delivered by sea or plane. Shipping the product would be ideal, the risk of detection is lower but would result in higher costs for a small amount of product. The product being delivered is only a test to check it's quality and will be the deciding factor in any decision to continue doing business with you. If the product is up to scratch, I will buy a larger supply next time. If we continue to do business it might be more practical to use shipping for future transactions.

This consignment will be delivered by plane, which will decrease travel time and lower the cost for all parties involved. Delivery by plane will bring along with it its own complications such as concealment, security screening/checks and drug dogs. I trust that you are aware of the risks and will take all necessary precautions.

Below is a map of drug flows from countries, could give us some info for later trades and where we can get more product from in the future.

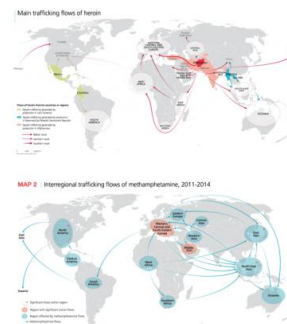


Figure 42: The file Memo containing method to potentially smuggle drugs

3.7.5. DISCUSSION FOR JANE ESTEBAN – EXCULPATORY

Jane's documents, like the undercover survival guide and possible briefing notes, support her connection to law enforcement rather than to drug dealing. They demonstrate preparation for a covert role and awareness of personal safety, but do not show her making money, running clients or planning shipments.

3.7.6. SUPPORTING EVIDENCE – JANE ESTEBAN

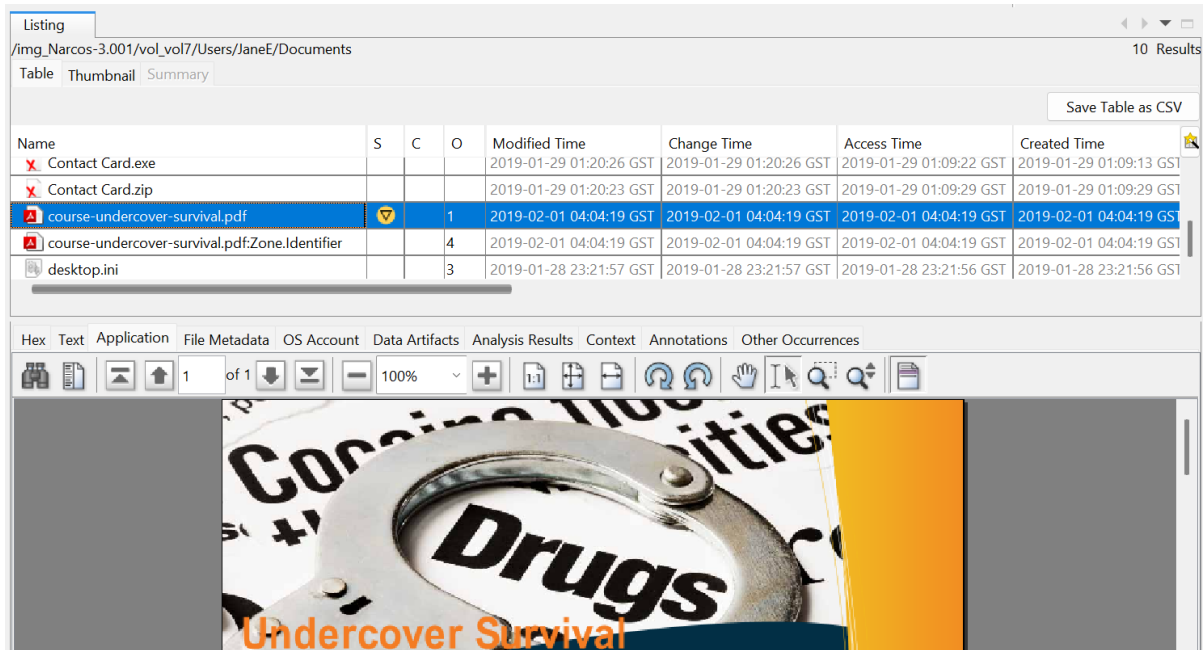


Figure 43: Document - undercover survival guide found in Jane's laptop

3.8. FINDINGS 8 – Identify any obfuscation methods used by the suspects.

3.8.1. DISCUSSION FOR STEVE KOWHAI

Steve uses several clear obfuscation techniques. The installation of CCleaner indicates that he deletes browsing history and other artefacts deliberately. The use of steganography to hide data within images and reliance on secure email further shows that he is actively trying to conceal content and communications.

3.8.2. SUPPORTING EVIDENCE – STEVE KOWHAI

Metadata	
Name:	/img_Narcos-1.001/vol_vol7/Program Files/CCleaner/CCleaner.exe
Type:	File System
MIME Type:	application/x-dosexec
Size:	14679256
File Name Allocation:	Allocated
Metadata Allocation:	Allocated
Modified:	2019-01-10 14:01:44 GST
Accessed:	2019-02-02 06:38:42 GST
Created:	2019-01-10 14:01:44 GST
Changed:	2019-01-30 03:41:21 GST
MD5:	f0f196fd71b9c7d65929861dd1bcfdcf
SHA-256:	00564fbe89d79ba9aa80a8c2a11c0f721e5ffda0ca0549273a22cf270e52664f
Hash Lookup Results:	UNKNOWN
Internal ID:	30207

Figure 44: Use of CCleaner

3.8.3. DISCUSSION FOR JOHN FREDRICKSON – INCULPATORY

John's main obfuscation method is the use of encryption. The TrueCrypt container holding the memo is stored under a folder labelled "important" but hidden behind a password. He also follows Steve's suggestions for stego-based communication.

3.8.4. SUPPORTING EVIDENCE – JOHN FREDRICKSON

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
Attachments-Important, crucial to our method				2019-01-30 04:00:49 GST	2019-01-30 04:00:50 GST	2019-02-02 06:30:38 GST	2019-01-30 04:00:48 GST	144	Allocated	Allocated	unknown
Contact_Card				2019-01-30 02:03:04 GST	2019-01-30 02:03:04 GST	2019-01-30 11:42:52 GST	2019-01-30 02:03:04 GST	280	Allocated	Allocated	unknown
Apache_OpenOffice_4.1.6_Win_x86_install_en-US.exe	1			2019-01-30 00:21:27 GST	2019-01-30 00:22:56 GST	2019-01-30 00:26:20 GST	2019-01-29 00:20:17 GST	141336318	Allocated	Allocated	unknown
Attachments-Important, crucial to our method.zip	0			2019-01-30 04:00:16 GST	2019-01-30 04:00:16 GST	2019-01-30 04:00:50 GST	2019-01-30 04:00:16 GST	5242990	Allocated	Allocated	unknown
Attachments-Important, crucial to our method.zip	0			2019-01-30 04:00:16 GST	2019-01-30 04:00:16 GST	2019-01-30 04:00:50 GST	2019-01-30 04:00:16 GST	241	Allocated	Allocated	unknown
BayPro_Setup_Mini_C1.exe	0			2019-01-29 00:17:25 GST	2019-01-29 00:17:56 GST	2019-01-29 00:20:22 GST	2019-01-29 00:17:24 GST	2033328	Allocated	Allocated	unknown
Contact_Card.zip	2			2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-30 02:03:04 GST	2019-01-30 02:01:31 GST	198604	Allocated	Allocated	unknown
Contact_Card.zip:Zone.Identifier	0			2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-30 02:03:04 GST	2019-01-30 02:01:31 GST	129	Allocated	Allocated	unknown
desktop.ini	3			2019-01-30 11:42:16 GST	2019-01-30 11:42:16 GST	2019-02-02 07:01:17 GST	2019-01-28 23:26:35 GST	282	Allocated	Allocated	unknown
DiscordSetup.exe	1			2019-01-29 00:21:04 GST	2019-01-29 00:21:04 GST	2019-01-29 00:25:10 GST	2019-01-29 00:21:01 GST	61394264	Allocated	Allocated	unknown
DiscordSetup.exe:Zone.Identifier	2			2019-01-29 00:21:04 GST	2019-01-29 00:21:04 GST	2019-01-29 00:25:10 GST	2019-01-29 00:21:01 GST	26	Allocated	Allocated	unknown
Image Steganography 1.5.2 Setup.exe	1			2019-02-01 03:00:17 GST	2019-02-01 03:00:43 GST	2019-02-02 06:48:15 GST	2019-02-01 03:00:15 GST	539616	Allocated	Allocated	unknown
TrueCrypt Setup 7.1a.exe	2			2019-01-30 03:58:55 GST	2019-01-30 03:58:59 GST	2019-02-02 06:48:15 GST	2019-01-30 03:58:55 GST	3466248	Allocated	Allocated	unknown

Figure 45: Use of Encryption Tool

3.8.5. DISCUSSION FOR JANE ESTEBAN – EXCULPATORY

In Jane’s device there is no major use of system cleaning tools or personal encrypted containers. The lack of obfuscation on her device compared with John and Steve’s behavior supports that she was not trying to cover personal criminal actions.

3.9. FINDINGS 9 – Identify encryption methods used by the suspects and determine two methods that can circumvent the encryption.

3.9.1. DISCUSSION

In this case, TrueCrypt is the main encryption tool in use. John creates an encrypted container named “secret” and stores inside it a memo directly related to the smuggling process.

Password and key discovery through artefact analysis

Instead of brute forcing, investigators can examine browser history, notes, configuration files and chat logs to find passwords or hints. In this case, the password “ilovediving” was identified from surrounding artefacts, allowing direct access to the container.

3.9.2. SUPPORTING EVIDENCE

crayfish1980	Ya.. I just told you about the tool  Received it. Will check to see if it works and confirm soon.
heresjohnny1	It worked , sending it and the password via email now. I used a tool called image steganography.
heresjohnny1	Okay, I'll have a look and see if I can get it to work and then send the image through
crayfish1980	A way of hiding one image within another. There's a simple application called 'Image Steganography'.
heresjohnny1	No what's that?
crayfish1980	Good Thinking, I already know how. Heard of steganography?
heresjohnny1	I want to send an image an image of something to you but it needs to be done safely. Any ideas?

Figure 46: Discussion about using an image steganography tool

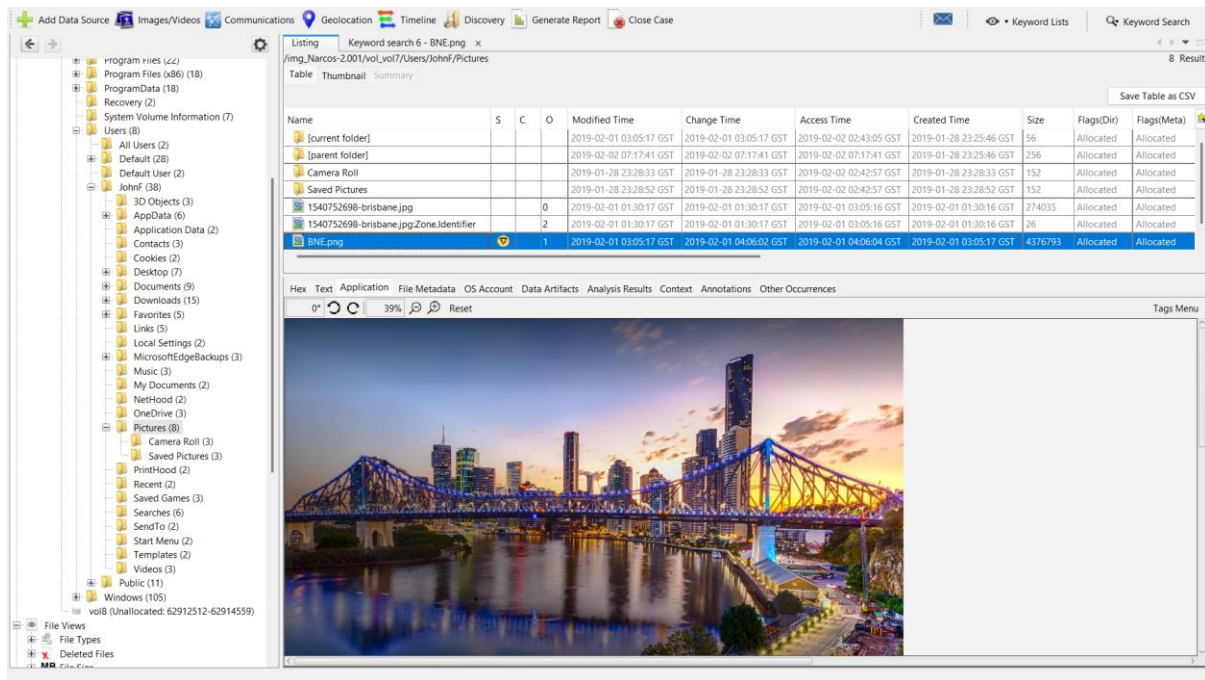


Figure 47: A picture named BNE.png found to have hidden message

[(4) Inbox | heresjohnny1@protonmail.com | ProtonMail - Mozilla Firefox - 10:05]
crayfish1980@protstego imageimage attached.[Enter]
pass is Elchapo2

Figure 48: Password found through logs

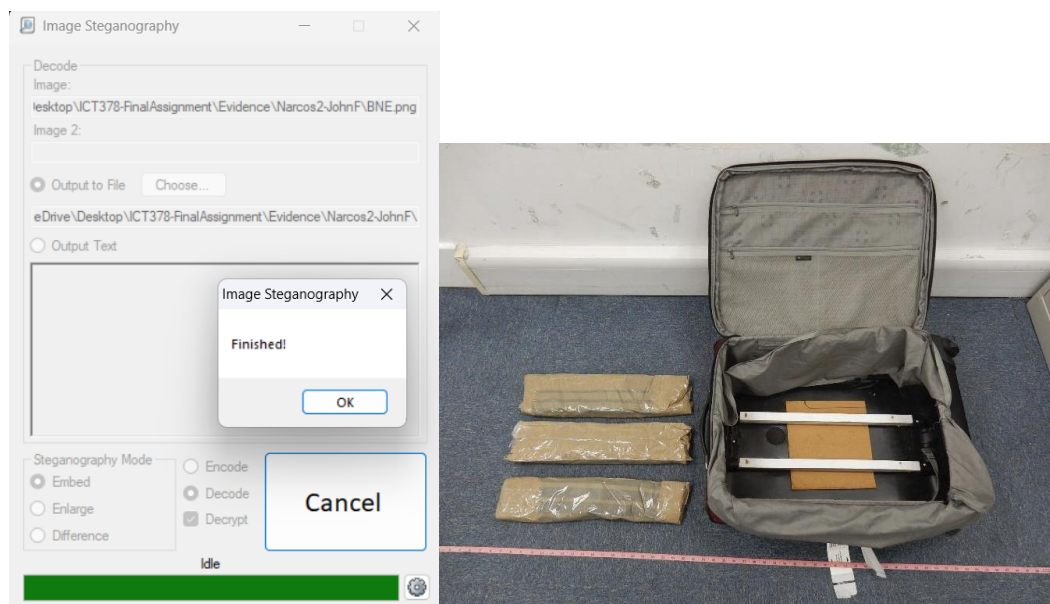


Figure 49: Image seen after decrypting

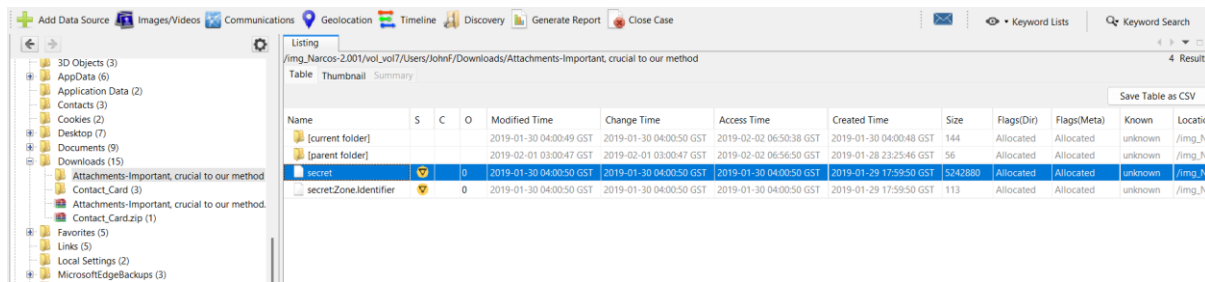


Figure 50: File named "secret" found in downloads under a folder "crucial to our method"

[Enter password for C:\Users\JohnF\Downloads\Attachments-Importa...\secret - 10:04]
ilovediving

Figure 51: Password found through system logs

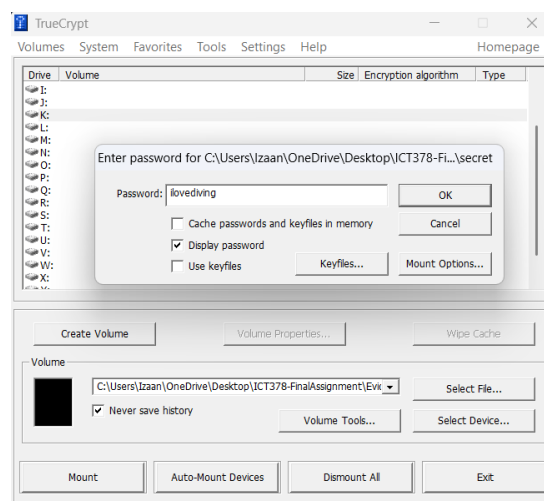


Figure 52: Decrypting it using TrueCrypt

Crystal Method

Destination – Wellington

Product – 1 kg crystal

Source – Brisbane (JF)

Wellington is part of the lower north island of New Zealand. It is accessible by road and sea. As tempting as the sea is, be aware that the coast of NZ may seem ideal, but everyone knows it's vulnerable as there have been several news reports about it.

The product is coming from Brisbane, Australia and as a result, will need to be delivered by sea or plane. Shipping the product would be ideal, the risk of detection is lower but would result in higher costs for a small amount of product. The product being delivered is only a test to check it's quality and will be the deciding factor in any decision to continue doing business with you. If the product is up to scratch, I will buy a larger supply next time. If we continue to do business it might be more practical to use shipping for future transactions.

This consignment will be delivered by plane, which will decrease travel time and lower the cost for all parties involved. Delivery by plane will bring along with it its own complications such as concealment, security screening/checks and drug dogs. I trust that you are aware of the risks and will take all necessary precautions.

Below is a map of drug flows from countries, could give us some info for later trades and where we can get more product from in the future.

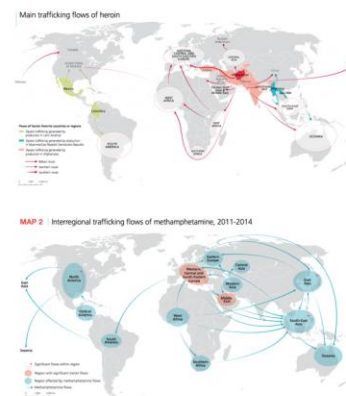


Figure 53: Memo file seen after decrypting

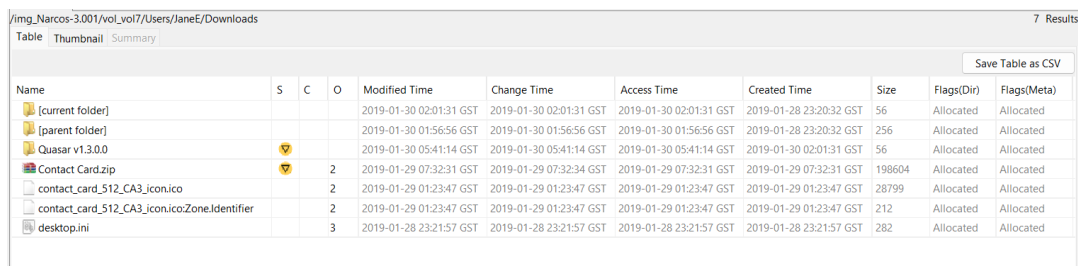
3.10. FINDINGS 10 – Identify malware used by one of the suspects and determine its purpose.

3.10.1. DISCUSSION

The malware identified is Quasar RAT, a remote access tool which can record keystrokes, take screenshots and monitor a target system. On Jane's laptop, the RAT files are in a bundle to be sent to John. On John's laptop, this bundle and its execution appear in downloads. Jane uses Quasar RAT as part of her undercover work to secretly observe John's activity and collect evidence. For John, however, the RAT becomes an unwanted infection that slows down his laptop and triggers his later search for antivirus software.

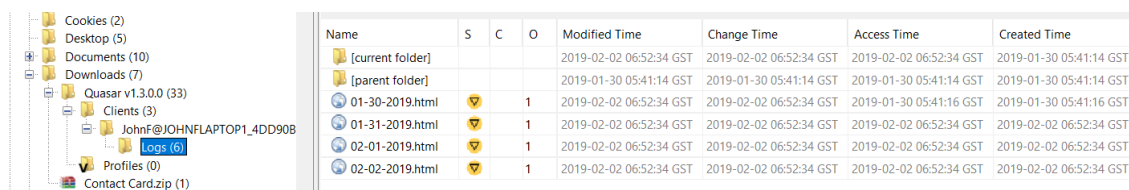
The purpose of the malware in this scenario is therefore covert surveillance of a suspect by an undercover officer, rather than financial gain or random infection. It still shows that John executed an unverified file, which shows poor security behaviour and allowed his device to be monitored.

3.10.2. SUPPORTING EVIDENCE



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
[current folder]				2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-28 23:20:32 GST	56	Allocated	Allocated
[parent folder]				2019-01-30 01:56:56 GST	2019-01-30 01:56:56 GST	2019-01-30 01:56:56 GST	2019-01-28 23:20:32 GST	256	Allocated	Allocated
Quasar v1.3.0.0				2019-01-30 05:41:14 GST	2019-01-30 05:41:14 GST	2019-01-30 05:41:14 GST	2019-01-30 02:01:31 GST	56	Allocated	Allocated
Contact Card.zip			2	2019-01-29 07:32:31 GST	2019-01-29 07:32:34 GST	2019-01-29 07:32:31 GST	2019-01-29 07:32:31 GST	198604	Allocated	Allocated
contact_card_512_CA3.ico			2	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	28799	Allocated	Allocated
contact_card_512_CA3.ico:ico:Zone.Identifier			2	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	2019-01-29 01:23:47 GST	212	Allocated	Allocated
desktop.ini			3	2019-01-28 23:21:57 GST	2019-01-28 23:21:57 GST	2019-01-28 23:21:57 GST	2019-01-28 23:21:57 GST	282	Allocated	Allocated

Figure 54: Downloads folder in Jane's laptop contain the Quasar RAT bundle, and a zip file named "contact card" which she used to send to John



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
[current folder]				2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-01-30 05:41:14 GST
[parent folder]				2019-01-30 05:41:14 GST	2019-01-30 05:41:14 GST	2019-01-30 05:41:14 GST	2019-01-30 05:41:14 GST
01-30-2019.html			1	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-01-30 05:41:16 GST	2019-01-30 05:41:16 GST
01-31-2019.html			1	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST
02-01-2019.html			1	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST
02-02-2019.html			1	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST	2019-02-02 06:52:34 GST

Figure 55: Logs were found under John's Laptop's name

[Enter password for C:\Users\JohnF\Downloads\Attachments-Importa...\secret - 10:04]
ilovediving

[Login | ProtonMail - Mozilla Firefox - 10:04]
crat[Back]yfi[Back]heresjohnny1@[Back]John1234

[4] Inbox | heresjohnny1@protonmail.com | ProtonMail - Mozilla Firefox - 10:05
crayfish1980@protstego imageimage attached.[Enter]
pass is Elchapo2

[Save As - 09:04]
BNE

[Enter Password - 09:05]
Elchapo2

Figure 56: Some important passwords obtained through logs

3.11. FINDINGS 11 – Identify the vulnerability that allowed the malware to function.

3.11.1. DISCUSSION

The key vulnerability in this case is social engineering and user action rather than a software bug. John received a file from Jane, believed it to be safe, and manually ran it. Because he ran the executable using his normal user privileges, the RAT was able to install and operate without needing an exploit. The lack of strong endpoint controls or strict security policy on his laptop also contributed, as there was nothing to block or isolate unknown executables.

From an incident-response perspective, this example shows how trust relationships and poor user awareness can be just as dangerous as technical flaws. Even if the operating system is fully patched, a user who runs a malicious program can still compromise the system.

3.11.2. SUPPORTING EVIDENCE

becomingjane	The usual
heresjohnny1	Also that contact card didn't open and I can't be bothered with new clients right now... Bigger fish to fry
heresjohnny1	Yo, what you after?
heresjohnny1	Sweet thanks
becomingjane	Also I've got some friends that want to score too. Here's their contact card
becomingjane	Got any of that ice??
becomingjane	

Figure 57: Jane mentions she has some friends interested and sends in "contact_card" (which had the virus)

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Met)
[current folder]				2019-02-01 03:00:47 GST	2019-02-01 03:00:47 GST	2019-02-02 06:56:50 GST	2019-01-28 23:25:46 GST	56	Allocated	Allocated
[parent folder]				2019-02-02 07:17:41 GST	2019-02-02 07:17:41 GST	2019-02-02 07:17:41 GST	2019-01-28 23:25:46 GST	256	Allocated	Allocated
Attachments-Important, crucial to our method				2019-01-30 04:00:49 GST	2019-01-30 04:00:50 GST	2019-02-02 06:50:38 GST	2019-01-30 04:00:48 GST	144	Allocated	Allocated
Contact_Card				2019-01-30 02:03:04 GST	2019-01-30 02:03:04 GST	2019-01-30 11:42:52 GST	2019-01-30 02:03:04 GST	280	Allocated	Allocated
Apache_OpenOffice_4.1.6_Win_x86_install_en-US.exe			1	2019-01-29 00:21:27 GST	2019-01-29 00:22:56 GST	2019-01-29 00:26:20 GST	2019-01-29 00:20:17 GST	141336318	Allocated	Allocated
Attachments-Important, crucial to our method.zip			0	2019-01-30 04:00:16 GST	2019-01-30 04:00:16 GST	2019-01-30 04:00:50 GST	2019-01-30 04:00:16 GST	5242990	Allocated	Allocated
Attachments-Important, crucial to our method.zip:Zone.Identifier			0	2019-01-30 04:00:16 GST	2019-01-30 04:00:16 GST	2019-01-30 04:00:50 GST	2019-01-30 04:00:16 GST	241	Allocated	Allocated
BavPro_Setup_Mini_C1.exe			0	2019-01-29 00:17:25 GST	2019-01-29 00:17:56 GST	2019-01-29 00:20:22 GST	2019-01-29 00:17:24 GST	2033328	Allocated	Allocated
Contact_Card.zip			2	2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-30 02:03:04 GST	2019-01-30 02:01:31 GST	198604	Allocated	Allocated
Contact_Card.zip:Zone.Identifier			0	2019-01-30 02:01:31 GST	2019-01-30 02:01:31 GST	2019-01-30 02:03:04 GST	2019-01-30 02:01:31 GST	129	Allocated	Allocated
desktop.ini			3	2019-01-30 11:42:16 GST	2019-01-30 11:42:16 GST	2019-02-02 07:01:17 GST	2019-01-28 23:26:35 GST	282	Allocated	Allocated
DiscordSetup.exe			1	2019-01-29 00:21:04 GST	2019-01-29 00:21:04 GST	2019-01-29 00:25:10 GST	2019-01-29 00:21:01 GST	61394264	Allocated	Allocated
DiscordSetup.exe:Zone.Identifier			2	2019-01-29 00:21:04 GST	2019-01-29 00:21:04 GST	2019-01-29 00:25:10 GST	2019-01-29 00:21:01 GST	26	Allocated	Allocated
Image Steganography 1.5.2 Setup.exe			1	2019-02-01 03:00:17 GST	2019-02-01 03:00:43 GST	2019-02-02 06:48:15 GST	2019-02-01 03:00:15 GST	559616	Allocated	Allocated
TrueCrypt Setup 7.1a.exe			2	2019-01-30 03:58:55 GST	2019-01-30 03:58:59 GST	2019-02-02 06:48:15 GST	2019-01-30 03:58:55 GST	3466248	Allocated	Allocated

Figure 58: John's downloads contained the "contact card" file indicating he downloaded and extracted it

Visit Details	
Title:	baidu antivirus download - Google Search
Date Accessed:	2019-01-29 00:16:05 GST
Domain:	google.com
URL:	https://www.google.com/search?client=firefox-b-ab&ei=_WJPXKO3Atr6rQGikbfwAw&q=baidu+antivirus+download&oq=baidu+antivirus+download&gs_l=psy-a
Referrer URL:	https://www.google.com/search?client=firefox-b-ab&q=badiu+anti-virus
Program Name:	Firefox Analyzer
Source	
Host:	Narcos-2.001_1 Host
Data Source:	Narcos-2.001
File:	/img/Narcos-2.001/vol_vol7/Users/JohnF/AppData/Roaming/Mozilla/Firefox/Profiles/tj248tbu.default/places.sqlite

Figure 59: John's history shows him searching for an anti-virus called baidu anti-virus, indicating that he suspects his system has been exploited

3.12. FINDINGS 12 – Leverage other Windows artefacts that provide corroborative evidence such as Windows Timeliner.

3.12.1. DISCUSSION

A range of Windows artefacts such as prefetch files, recent document lists, registry keys, event logs and timeline/activity history were examined to support the evidence already gathered from files and chats.

Using these artefacts strengthens the case by showing that the sequence of events is consistent across different sources. It also helps rule out claims that files were placed on the systems without the suspects ever opening or using them.

3.12.2. SUPPORTING EVIDENCE

ActivityId	Platform	PackageName	ExpirationTime
706	BLOB	windows_universal	1551572608
707	BLOB	msa	1551572608
708	BLOB	msa	1551572608
709	BLOB	android	1551572608
710	BLOB	android	1551572608
711	BLOB	android	1551572608
712	BLOB	ios	1551572608
713	BLOB	ios	1551572608
714	BLOB	ios	1551572608
715	BLOB	msa	1551572608
716	BLOB	afs_crossplatform	1551572608
717	BLOB	windows_win32	1551572608
718	BLOB	packageid	1551572608
719	BLOB	host	1551572608
720	BLOB	afs_crossplatform	1551832825
721	BLOB	windows_win32	1551832825
722	BLOB	packageid	1551832825
723	BLOB	host	1551832825
724	BLOB	afs_crossplatform	1551572608
725	BLOB	windows_win32	1551572608
726	BLOB	packageid	1551572608
727	BLOB	host	1551572608
728	BLOB	host	1551572608
729	BLOB	afs_crossplatform	1551572608
730	BLOB	windows_universal	1551572608

Figure 60: Extract from Steve's ActivitiesCache.db showing Windows Timeline application activity, including repeated Microsoft Edge usage and package-based execution records.

Database Structure | Browse Data | Edit Pragma | Execute SQL

Activity | Activity_PackageId

Table: Activity_PackageId

ActivityId	Platform	PackageName	ExpirationTime
103	BLOB	x_exe_path	c:\...
104	BLOB	host	c:\...
105	BLOB	x_exe_path	c:\...
106	BLOB	packageid	c:\...
107	BLOB	host	c:\...
108	BLOB	windows_win32	com.squirrel.discord.discord
109	BLOB	host	com.squirrel.discord.discord
110	BLOB	windows_win32	com.squirrel.discord.discord
111	BLOB	packageid	com.squirrel.discord.discord
112	BLOB	host	com.squirrel.discord.discord
113	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)
114	BLOB	host	(d65231b0-b2f1-4857-a4ce-...)
115	BLOB	windows_win32	com.squirrel.discord.discord
116	BLOB	packageid	com.squirrel.discord.discord
117	BLOB	host	com.squirrel.discord.discord
118	BLOB	x_exe_path	c:\...
119	BLOB	host	c:\...
120	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)
121	BLOB	host	(d65231b0-b2f1-4857-a4ce-...)
122	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)
123	BLOB	packageid	(d65231b0-b2f1-4857-a4ce-...)
124	BLOB	host	(d65231b0-b2f1-4857-a4ce-...)
125	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)
126	BLOB	packageid	(d65231b0-b2f1-4857-a4ce-...)
127	BLOB	host	(d65231b0-b2f1-4857-a4ce-...)

Go to: 1

Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)

Remote Identity Select an identity to connect

DBHub.io Local Current Database

Name Last modified Size

SQL Log Plot DB Schema Remote

Figure 61: Extract from John's ActivitiesCache.db illustrating recorded application execution events used to validate his browsing and program activity timeline.

Directory Tree

Listing

Recent Documents

Source Name	S	C	O	Path	Date Accessed	Data Source	Comment
The Internet.Ink				No preferred path found	2019-01-28 23:26:49 GST	Narcos-2.001	
Network and Internet.Ink				No preferred path found	2019-01-28 23:54:18 GST	Narcos-2.001	
Network and Sharing Center.Ink				No preferred path found	2019-01-28 23:54:18 GST	Narcos-2.001	
Local Disk (E:).Ink				E:\	2019-01-30 04:04:22 GST	Narcos-2.001	
CD Drive.Ink				D:\	2019-01-28 23:30:07 GST	Narcos-2.001	
No preferred path found.Ink				No preferred path found	0000-00-00 00:00:00	Narcos-2.001	
Ink				E\	0000-00-00 00:00:00	Narcos-2.001	
autorun.Ink				D:\autorun.ico	2019-01-28 23:30:07 GST	Narcos-2.001	
Memo Things.Ink				E:\Memo Things.odt	2019-01-30 04:04:22 GST	Narcos-2.001	
Memo Things.odt				E:\Memo Things.odt	0000-00-00 00:00:00	Narcos-2.001	
autorun.ico				D:\autorun.ico	0000-00-00 00:00:00	Narcos-2.001	
Pictures.Ink				C:\Users\John\F\Pictures	2019-02-01 01:30:17 GST	Narcos-2.001	
Contact_Card.Ink				C:\Users\John\F\Downloads>Contact_Card.zip	2019-01-30 02:01:31 GST	Narcos-2.001	
Music.Ink				C:\Users\John\F\Music	0000-00-00 00:00:00	Narcos-2.001	
Videos.Ink				C:\Users\John\F\Videos	0000-00-00 00:00:00	Narcos-2.001	
Desktop.Ink				C:\Users\John\F\Desktop	0000-00-00 00:00:00	Narcos-2.001	

Save Table as CSV

Figure 62: Recent Files shows recently accessed files

3.13. FINDINGS 13 – Identify whether changes have occurred to these artefacts across the different operating system builds.

3.13.1. DISCUSSION

The three Narcos systems run different builds of Windows-10, and this has an important effect on forensic work.

During the investigation, these differences were accounted for by checking version numbers and adjusting search paths appropriately.

3.13.2. SUPPORTING EVIDENCE

ActivityId	Platform	PackageName	ExpirationTime
706	windows_universal	microsoft.microsoftedge_9wekyb3d8bbwe	1551572608
707	msa	f44b1140-bc5e-48c6-9dc0-5cfa53c0e34	1551572608
708	msa	000000004c1bc462	1551572608
709	android	com.microsoft.emmx.daily	1551572608
710	android	com.microsoft.emmx.selfhost	1551572608
711	android	com.microsoft.emmx	1551572608
712	ios	com.microsoft.emmx.daily-df	1551572608
713	ios	com.microsoft.emmx.selfhost-df	1551572608
714	ios	com.microsoft.msedge	1551572608
715	msa	2710a4d6-9460-4830-9314-189867ff160f	1551572608
716	afs_crossplatform	4677216866307413433	1551572608
717	windows_win32	microsoft.windows.explorer	1551572608
718	packageid	microsoft.windows.explorer	1551572608
719	host		1551572608
720	afs_crossplatform	15241238769019164279	1551832825
721	windows_win32	com.squirrel.discord.discord	1551832825
722	packageid	com.squirrel.discord.discord	1551832825
723	host		1551832825
724	afs_crossplatform	15241238769019164279	1551572608
725	windows_win32	com.squirrel.discord.discord	1551572608
726	packageid	com.squirrel.discord.discord	1551572608
727	host		1551572608
728	host	edge.activity.windows.com	1551572608
729	afs_crossplatform	11657492638432899311	1551572608
730	windows_universal	microsoft.microsoftedge_9wekyb3d8bbwe	1551572608

Figure 63: Steve's ActivitiesCache.db (Activity_PackageId) showing multiple Timeline entries

Database StructureBrowse DataEdit PragmaExecute SQL

ActivityActivity_PackageId

Activity_PackageId

Table: Activity_PackageId

	ActivityId	Platform	PackageId	ExpirationTime
	Filter	Filter	Filter	Filter
103	BLOB	x_exe_path	c:...	1551298887
104	BLOB	host	c:...	1551298887
105	BLOB	x_exe_path	c:...	1551298889
106	BLOB	packageid	c:...	1551298889
107	BLOB	host	c:...	1551298889
108	BLOB	windows_win32	com.squirrel.discord.discord	1551298899
109	BLOB	host	c:...	1551298899
110	BLOB	windows_win32	com.squirrel.discord.discord	1551298902
111	BLOB	packageid	com.squirrel.discord.discord	1551298902
112	BLOB	host	c:...	1551298902
113	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)	1551298902
114	BLOB	host	c:...	1551298902
115	BLOB	windows_win32	com.squirrel.discord.discord	1551298916
116	BLOB	packageid	com.squirrel.discord.discord	1551298916
117	BLOB	host	c:...	1551298916
118	BLOB	x_exe_path	c:...	1551298980
119	BLOB	host	c:...	1551298980
120	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)	1551298989
121	BLOB	host	c:...	1551298989
122	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)	1551298992
123	BLOB	packageid	(d65231b0-b2f1-4857-a4ce-...)	1551298992
124	BLOB	host	c:...	1551298992
125	BLOB	x_exe_path	(d65231b0-b2f1-4857-a4ce-...)	1551298999
126	BLOB	packageid	(d65231b0-b2f1-4857-a4ce-...)	1551298999
127	BLOB	host	c:...	1551298999

Go to: 1

Edit Database Cell

Mode: Text

Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)

Remote

Identity Select an identity to connect

DBHubLocalCurrent Database

Name	Last modified	Size
------	---------------	------

Figure 64: John's ActivitiesCache.db (Activity_PackageId) showing fewer recorded entries.

The screenshot shows a database browser window with the table 'Activity' selected. The table has many columns: Id, AppId, AppActivityId, ActivityType, ActivityStatus, ParentActivityId, Tag, Group, MatchId, LastModifiedTime, ExpirationTime, Payload, Priority, OriginatingDevice, IsLocalOnly, and Platform. The data is filtered to show rows 0 to 0. The 'ExpirationTime' column contains values ranging from 1551298887 to 1551298999. The 'Platform' column contains values like 'x_exe_path', 'host', and 'windows_win32'. The 'ActivityId' column contains values like 'BLOB' and 'packageid'.

Id	AppId	AppActivityId	ActivityType	ActivityStatus	ParentActivityId	Tag	Group	MatchId	LastModifiedTime	ExpirationTime	Payload	Priority	OriginatingDevice	IsLocalOnly	Platform
0															

Figure 65: Jane's ActivitiesCache.db showing an empty Browsing tab, indicating that her Windows build did not generate Timeline records.

4. CONCLUSION

4.1. SUMMARY OF FINDINGS

The analysis of the three datasets - Narcos 1 (Steve), Narcos 2 (John), and Narcos 3 (Jane) - included a detailed review of drive images, memory dumps, browser history, Discord traces, and system artefacts. The investigation confirmed signs of methamphetamine-related activity on two of the devices. Key artefacts such as communication logs, deleted files, search activity, and system records clearly supported the involvement of Steve and John in drug-related planning. Jane's device contained minimal artefacts and showed no indication of illegal intent or behaviour.

4.2. EVIDENCE OF ILLEGAL DRUG ACTIVITY

Evidence across Steve's and John's devices strongly supports involvement in methamphetamine activity. This included drug-related web searches, communication about "samples," encrypted documents, steganography usage, and browser history indicating preparation and coordination. These artefacts collectively demonstrate clear knowledge and intent relating to drug activity. No such evidence was found on Jane's device.

4.3. RELIABILITY OF EVIDENCE COLLECTION

The collection and analysis process is reliable. All images were examined in read-only mode, hash values were preserved, and tools such as Autopsy, Magnet AXIOM, and DB Browser were used according to accepted forensic procedures. Cross-checking artefacts between tools further supports the accuracy of the findings.

4.4. INNOCENT / GUILTY

- **Steve Kowhai – Guilty**

His system shows drug-cutting research, communication with John, encrypted files, wiping tools, and activity tied to the house where drugs were found.

- **John Fredricksen – Guilty**

His device contains drug-related searches, deleted files, Discord remnants, and evidence of planning delivery with Steve.

- **Jane Esteban – Innocent**

Her device contains almost no incriminating artefacts, no relevant searches, and no activity linking her to drug planning.

5. APPENDIX

5.1. PERSONS OF INTEREST

Person	Description
Steve Kowhai	Resident of Wellington and owner of the property at 666 Rewera Avenue. Displays technical knowledge, researched crystal meth preparation, communicated with John, used steganography and encryption, and attempted to wipe evidence using CCleaner. Strong involvement in methamphetamine operations.
John Fredricksen	Communicated frequently with Steve regarding drug delivery. Travelled into New Zealand carrying suspicious items. His device contained searches, deleted files, Discord remnants, and documents linked to drug handling. Clear involvement in smuggling coordination.
Jane Esteban	Minimal digital activity found. No drug-related searches or messages. Her artefacts and OS timeline were sparse, suggesting no involvement. Based on digital evidence, no strong indicators of criminal participation were found.

5.2. ASSOCIATION DIAGRAM

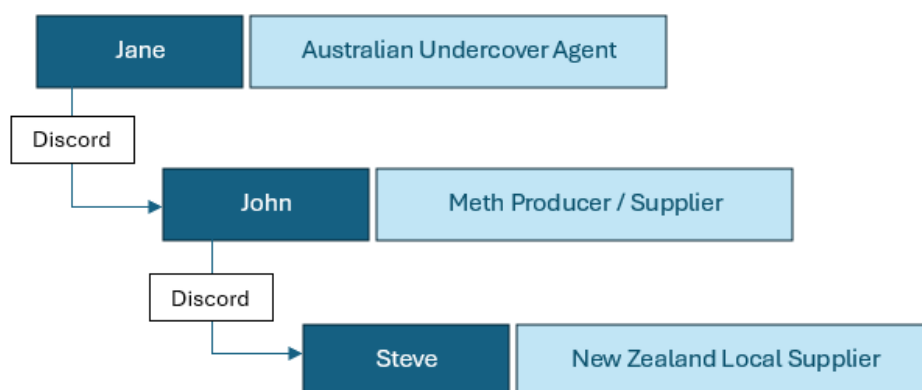


Figure 66: Association Diagram

5.3. TABLE OF FINDINGS

Name	Description	Directory
Memo Things.odt	Contains the Crystal method	C://Users/JohnF/Documents/Business/Attachments-Crucial to our method/secret
Secret	Contains Memo Things.ODT which is locked using TrueCrypt	C://Users/JohnF/Documents/Business/Attachments-Crucial to our method/secret
Method run.jpg	screenshot of directions from/to Eastborn Library saved as "Method run"	C://Users/Steve/Documents/Misc/Method run.jpg
airportcrystals.jpg	showing directions to a certain location	C://Users/Steve/Documents/Misc/airport crystals.jpg
dropoff.png	showing a location - Eastborn Library, saved as "dropoff"	C://Users/Steve/Documents/Misc/dropoff.png
package.jpg	Contains John's Narcotics Pack	C://Users/Steve/Documents/Misc/BNE.png
Clients.ods	List of clients of John with their locations, amount, and delivery	C://Users/JohnF/Documents/Business/
Contact Card.zip	Quasar RAT used to infiltrate John	C://Users/JaneE/Downloads/
620x349.jpg	Picture crystal meth packages found in Steves Laptop	C://Users/Steve/Pictures/620x349.jpg
eight_col_patches_crp.jpg	Picture of a gang sign saved in steves Laptop	C://Users/Steve/Pictures/eight_col_patches_crp.jpg
price_meth_bust_4.jpg	Picture of money and drugs busted saved in Steves Laptop	C://Users/Steve/Pictures/price_meth_bust-4.jpg
Steve K.PNG	Screenshot of a flight ticket from Brisbane to Wellington	C://Users/JohnF/Documents/Business/Steve K.PNG
shipping.png	Screenshot of shipping details	C://Users/JohnF/Documents/Business/shipping.PNG
course-undercover-survival.pdf	A document on how to survive as an undercover	C://Users/JaneE/Documents/course-undercover-survival.pdf
Quasar v 1.3.0.0	Quasar RAT installed to create a virus	C://Users/JaneE/Downloads/Quasar v 1.3.0.0
afp.png	Australian Federal Police Badge	C://Users/JaneE/Pictures/afp.png
crysl.jpg	Crystal Meth pic saved in pictures folder	C://Users/JaneE/Pictures/crysl.jpg

5.4. EVIDENCE LISTING

- User accounts, computer names, and OS information
- Browser activity and drug-related searches
- Images that show behaviour patterns
- Binary files such as executables, encrypted containers, and deleted artefacts
- Means of communication (Discord traces, email remnants, browser history)
- Documents related to planning, travel, or concealed information
- Obfuscation methods (CCleaner, TrueCrypt use, steganography)
- Encryption methods and potential ways to bypass them
- Memory artefacts showing running processes and searches
- Windows Timeline artefacts showing differences across OS builds
- Roles of each suspect derived from communication and behaviour patterns

5.5. EVIDENCE TIMELINE

January 28th 2019 – First Discord chat between Steve and John

January 29th 2019 – Jane begins her investigation

January 29th 2019 – John contacts Steve telling him about a new supplier

January 29th 2019 – Jane creates contact with John acting like a meth head

January 29th 2019 – Steve makes secret document for smuggling

January 29th 2019 – Jane sends the “Contact Card” file containing the Quasar RAT

January 30th 2019 – Jane accesses John’s log files through the RAT she made and sent

January 30th 2019 – Steve sends the secret encrypted file to John through protonmail

January 30th 2019 – John blackmails Jane into coming him into New Zealand

January 30th 2019 – John makes web searches regarding drug concealment techniques

January 31th 2019 – Concealed Drugs Suitcase

January 31st 2019 – Drugs Images stored in recycle bin

February 1st 2019 – John sends images of drugs via Steganography Tool to Steve

February 2nd 2019 – John and Jane purchases tickets from Brisbane to Wellington

February 2nd 2019 – John and Jane get caught up by New Zealand customs.

5.6. SOFTWARE & TOOLS USED

- Autopsy
- Magnet AXIOM
- FTK Imager
- ChromeCacheView
- TrueCrypt
- Image Steganography Tool
- DB Browser for SQLite

5.7. REFERENCES

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